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ABSTRACT

In hypersonic computational fluid dynamics, the two-temperature (2-T) model is widely used to simulate thermochemical nonequilibrium. The 2-T model incorporates translational-rotational and electron-electronic-vibrational energies, assuming that the integrated energies have the equivalent temperature. In this study, multi-T models are constructed to accurately predict the effects on heat flux and free electrons due to the separation of energy modes under hypersonic environments. The three-temperature (3-T) model separates the electron-electronic energy from the electron-electronic-vibrational energy of the 2-T model. The 3-T model can accurately predict the distribution and temperature of free electrons by separating the energy of free electrons, which has different characteristics from heavy particles. The four-temperature model treats rotational energy as a nonequilibrium energy mode, distinct from translational-rotational energy. While the rotational temperature reaches equilibrium rapidly at low temperatures, at high-temperature regime rotational temperature shows a relaxation time similar to that of vibrational temperature, which cannot be ignored. To develop multi-T models, electron-vibrational relaxation and translational-rotational relaxation, which are omitted in the 2-T model, are considered. Various flight test and ground facility conditions are analyzed to verify the effects of electron and heat flux under circumstances that include shock, expansion, and shock wave boundary layer interaction. The results of the multi-T models show significant differences in electron temperature and distribution caused by electron-electronic nonequilibrium. Additionally, rotational nonequilibrium increases the shock standoff distance and alters the electron distribution at high altitudes. The heat flux difference across multi-T models is found to be negligible, except in the high degree of ionization condition.

I. INTRODUCTION

Hypersonic flow is characterized by extremely high temperatures and pressures of the gas due to strong shock waves. This flow characteristic affects not only the aerodynamic properties of hypersonic vehicles, but also the structural failure due to heated gas, making accurate prediction of flow phenomena essential. Downstream of the shock wave, phenomena such as dissociation reactions, ionization reactions, radiation, and thermal relaxation dominate. Most hypersonic computational fluid dynamics (CFD) simulations use the two-temperature (2-T) model by Park[1,2] to simulate this thermochemical nonequilibrium. Park's 2-T model separates the energy of the mixture into two energy modes by incorporating translational-rotational energy and electron-electronic-vibrational energy. Although this integration of energy modes was constrained by the limitations of the kinetic model at the time,[1] the dissociation reaction rate calculated by geometric averaging of the temperatures resulted in accurate heat flux predictions in most hypersonic environments.

An accurate prediction of flow phenomena is also essential in designing an appropriate thermal protection system (TPS) for hypersonic vehicles. Conventional TPSs utilize the phenomenon of ablation to reduce the heat load on hypersonic vehicles. This approach requires an accurate heat flux calculation to predict the surface temperature as the performance of the TPS is directly related to the temperature of the surface material. While this is an efficient way to reduce the surface heat load, a damaged heat shield cannot be reused, contrary to the goal of a reusable hypersonic system, which is being actively studied. A recent approach to reusable TPS is to use free electrons and electromagnetic forces. Free electrons have been studied as a cause of communication blackout in hypersonic communities, and have been studied to predict and mitigate the occurrence of such blackout.[3–7] Recent methods for reusable TPS include electron transpiration cooling[8–10] using thermionic emission and a magnetohydrodynamic enhanced system.[10–12] The magnetohydrodynamic enhanced system has been studied because of its possibility to control thermal behavior.[13] Since these approaches account for the energy state of free electrons and the strength of electromagnetic forces, an accurate prediction of the energy state of electrons as well as electron density is important for heat flux mitigation.

The 2-T model's accurate prediction of electron density has been confirmed by various studies. However, its prediction of the energy state of free electrons is inherently limited by the assumption that it uses electron-electronic-vibrational energy. This is because free electrons have extremely light mass and different electrical properties compared to heavy particles. Therefore, separation of electron energy is necessary to predict precise electron behavior. By separating the energy of free electrons under the continuum assumption, governing equations that can be applied to CFD solver were derived. Appleton and Bray[14] developed the conservation equation of plasma by considering the ionization of atoms and separating the temperature of heavy particles and electrons. Subsequently, Lee[15,16] proposed the governing equations separating the electron-electronic energy as a third energy mode and the relaxation time of the electron impact vibrational ($e-vib$) excitation process of nitrogen, which is applicable to large scale CFD simulations. Laporta *et al.*[17–19] additionally presented $e-vib$ relaxation times for nitrogen, oxygen, and nitric oxide. After the $e-vib$ energy relaxation times were proposed, CFD simulations were performed in various hypersonic environments applying the thermochemical nonequilibrium three-temperature (3-T) model.[20–24] Despite the implementation of the 3-T model in the CFD simulations, studies mainly focused on the $e-vib$ relaxation of nitrogen.

Most of the free electrons around a hypersonic vehicle are generated by the strong shock wave at the nose.[25] However, as the rotational nonequilibrium is known to increase the shock standoff distance,[2,26] the effect of electrons due to the change in shock wave structure at high altitude cannot be ignored. To accurately predict the distribution of free electrons, it is essential to consider rotational relaxation. For the translational-rotational $(trans-rot)$ relaxation of nitrogen, the effective rotational collision number was proposed by Parker[27] and Lordi and Mates[28] and was validated with experimental data[29,30] at temperatures below 1500 K. In the low temperature regime, the rotational temperature equilibrates rapidly, confirming that the translational-rotational temperature of the 2-T model is an appropriate assumption. However, since the rotational relaxation time of air surpasses the vibrational relaxation time above 12 000 K, it is an important factor in determining thermal nonequilibrium at high temperatures.[31] Subsequently, relaxation time models considering the rotational state-to-state transition of nitrogen have been developed in various studies.[26,31,32] However, the influence of rotational relaxation on free electron and surface heat flux in large scale CFD simulations has not been studied properly in detail. Consequently, thermochemical nonequilibrium four-temperature (4-T) is required to accurately predict the behavior of electrons with rotational relaxation.

This study investigates the effect of separating of energy modes on free electron and surface heat flux in a hypersonic flow field. The multi-T models (2-T, 3-T, 4-T) are developed with the separation of rotational energy and electron-electronic energy, which are integrated in the existing 2-T model. The multi-T models are applied to the CFD solver to perform numerical simulations under various ground facility and flight test conditions. To the best of the authors' knowledge, this study is the first attempt to develop thermochemical nonequilibrium models incorporating the latest thermochemical energy transfers, which have not been previously considered. By comparing the heat transfer and electron behavior across the multi-T models, the distinct characteristics of each thermal nonequilibrium model are expected to be clearly identified.

II. METHODOLOGY

A. Governing equation of multi-temperature models

The multi-T models begin with the separation of translational-rotational and electron-electronic-vibrational energy, which are the main components of the existing 2-T model.[33] The 2-T model reflects the thermochemical nonequilibrium of air by integrating energy modes with similar characteristics within the energies considered in the mixture for efficient analysis. Within the integrated energy, it assumes the equivalent temperature of the different energy modes. The extension to 3-T and 4-T means the separation of energy mode (temperature), and the properties of the separated energy modes can be captured precisely by applying an appropriate energy transfer model. Energy modes in the multi-T models exchange energy through interactions with other energy modes. Since the multi-T models do not affect the form of the species mass, momentum, and energy equations in the governing equations, these equations are identical for 2-T, 3-T, and 4-T models. The species mass equation for the multi-T models is

$$\frac{\partial}{\partial t}(\rho_s)+\frac{\partial}{\partial x}(\rho_s u)+\frac{\partial}{\partial y}(\rho_s v)=\frac{\partial}{\partial x}(-J_{s,x})+\frac{\partial}{\partial y}(-J_{s,y})+\dot{\omega}_s. \quad (1)$$

In Eq. (1), the subscript s represents the 11 chemical species of air considered in the governing equations: $(N_2, O_2, NO, N, O, N_2^+, O_2^+, NO^+, N^+, O^+, e^-)$. ρ_s, J_s, and $\dot{\omega}_s$ are the partial density, diffusion velocity, and production rate by chemical reactions for species s, respectively. The axisymmetric momentum equations for the mixture, including all chemical species, are

$$\frac{\partial}{\partial t}(\rho u)+\frac{\partial}{\partial x}(\rho uu+p)+\frac{\partial}{\partial y}(\rho uv)=\frac{\partial}{\partial x}(\tau_{xx})+\frac{\partial}{\partial y}(\tau_{xy}), \quad (2)$$

$$\frac{\partial}{\partial t}(\rho v) + \frac{\partial}{\partial x}(\rho u v) + \frac{\partial}{\partial y}(\rho v v + p) = \frac{\partial}{\partial x}(\tau_{xy}) + \frac{\partial}{\partial y}(\tau_{yy}) + \frac{\alpha}{y_c}(-\tau_{\theta\theta} + p). \tag{3}$$

In Eqs. (2) and (3), ρ, u, v, p, and τ represent the density of the mixture, velocity in the x and y directions, pressure, and the viscous stress tensor, respectively. α is an axisymmetric switch, set to 1 under the assumption of axisymmetry and 0 otherwise. The energy equation for the mixture is

$$\frac{\partial}{\partial t}(\rho e) + \frac{\partial}{\partial x}(\rho h u) + \frac{\partial}{\partial y}(\rho h v) = \frac{\partial}{\partial x}(u\tau_{xx} + v\tau_{xy} - q_x) + \frac{\partial}{\partial y}(u\tau_{xy} + v\tau_{yy} - q_y). \tag{4}$$

In Eq. (4), e, h, and q are the total specific energy, total specific enthalpy, and energy diffusion by viscosity of the mixture, respectively. The nonequilibrium energy equations for the multi-T models used in this study are as follows.

1. 2-T: Electron-electronic-vibrational nonequilibrium model

The 2-T model assumes two energy modes: translational-rotational energy and electron-electronic-vibrational energy. The rotational energy combined with the translational energy is implicitly represented in Eq. (4). The 2-T electron-electronic-vibrational energy equation from Kim and Jo[33] is

$$\frac{\partial}{\partial t}(\rho e_{eev}) + \frac{\partial}{\partial x}(\rho h_{eev} u) + \frac{\partial}{\partial y}(\rho h_{eev} v) = \frac{\partial}{\partial x}(-q_{eev,x}) + \frac{\partial}{\partial y}(-q_{eev,y}) + S_{eev}, \tag{5}$$

$$S_{eev} = S_{trans-vib} - S_{vib-rot} + S_{trans-e} + S_{rot-e} + S_{chem-vib} + S_{chem-ee}. \tag{6}$$

In Eq. (5), e_{eev}, h_{eev}, q_{eev}, and S_{eev} represent the specific energy, specific enthalpy, diffusion term, and source term of the electron-electronic-vibrational energy mode, respectively. A detailed expression of S_{eev} is shown in Eq. (6).

2. 3-T: Vibrational, electron-electronic nonequilibrium model

The 3-T model was introduced by Lee,[15] who separated electron-electronic energy from electron-electronic-vibrational energy as the third energy mode. This form of energy integration can omit the electron impact electronic excitation energy transfer. Both vibrational and electron-electronic energy equations are

$$\frac{\partial}{\partial t}(\rho e_{vib}) + \frac{\partial}{\partial x}(\rho e_{vib} u) + \frac{\partial}{\partial y}(\rho e_{vib} v) = \frac{\partial}{\partial x}(-q_{vib,x}) + \frac{\partial}{\partial y}(-q_{vib,y}) + S_{vib}, \tag{7a}$$

$$\frac{\partial}{\partial t}(\rho e_{ee}) + \frac{\partial}{\partial x}(\rho h_{ee} u) + \frac{\partial}{\partial y}(\rho h_{ee} v) = \frac{\partial}{\partial x}(-q_{ee,x}) + \frac{\partial}{\partial y}(-q_{ee,y}) + S_{ee}, \tag{7b}$$

$$S_{vib} = S_{trans-vib} - S_{vib-rot} + S_{e-vib} + S_{chem-vib}, \tag{8a}$$

$$S_{ee} = S_{trans-e} + S_{rot-e} - S_{e-vib} + S_{chem-ee}. \tag{8b}$$

In Eq. (7a), e_{vib}, q_{vib}, and S_{vib} represent the specific energy, diffusion term, and source term of the vibrational energy mode, respectively. e_{ee}, h_{ee}, q_{ee}, and S_{ee} in Eq. (7b) are the specific energy, specific enthalpy, diffusion term, and source term of the electron-electronic energy mode, respectively. A detailed expression of S_{vib}, S_{ee} with additional $e - vib$ energy transfer is given by Eqs. (8a) and (8b).

3. 4-T: Rotational, vibrational, electron-electronic nonequilibrium model

The 4-T model separates rotational energy from the translational-rotational energy integrated in the 3-T model. The rotational nonequilibrium model by Jo *et al.*[26] is used to describe the rotational, vibrational, and electron-electronic energy equations as follows:

$$\frac{\partial}{\partial t}(\rho e_{rot}) + \frac{\partial}{\partial x}(\rho e_{rot} u) + \frac{\partial}{\partial y}(\rho e_{rot} v) = \frac{\partial}{\partial x}(-q_{rot,x}) + \frac{\partial}{\partial y}(-q_{rot,y}) + S_{rot}, \tag{9a}$$

$$\frac{\partial}{\partial t}(\rho e_{vib}) + \frac{\partial}{\partial x}(\rho e_{vib} u) + \frac{\partial}{\partial y}(\rho e_{vib} v) = \frac{\partial}{\partial x}(-q_{vib,x}) + \frac{\partial}{\partial y}(-q_{vib,y}) + S_{vib}, \tag{9b}$$

$$\frac{\partial}{\partial t}(\rho e_{ee}) + \frac{\partial}{\partial x}(\rho h_{ee} u) + \frac{\partial}{\partial y}(\rho h_{ee} v) = \frac{\partial}{\partial x}(-q_{ee,x}) + \frac{\partial}{\partial y}(-q_{ee,y}) + S_{ee}, \tag{9c}$$

$$S_{rot} = S_{trans-rot} + S_{vib-rot} - S_{rot-e} + S_{chem-rot}, \tag{10a}$$

$$S_{vib} = S_{trans-vib} - S_{vib-rot} + S_{e-vib} + S_{chem-vib}, \tag{10b}$$

$$S_{ee} = S_{trans-e} + S_{rot-e} - S_{e-vib} + S_{chem-ee}. \tag{10c}$$

In Eq. (9a), e_{rot}, q_{rot}, and S_{rot} represent the specific energy, diffusion term, and source term of the rotational energy mode, respectively. Detailed expressions of S_{rot}, S_{vib}, and S_{ee} with the additional $trans - rot$ energy transfer are given by Eq. (10).

B. Nonequilibrium source term of multi-temperature models

Each energy mode transfers energy through relaxation and chemical reactions. As shown in Eq. (10a), rotational energy undergoes energy relaxation with translational, vibrational, and electron-electronic energy by $S_{trans-rot}$, $S_{vib-rot}$, S_{rot-e}, and reflects rotational energy loss due to chemical reactions by $S_{chem-rot}$. Similarly, the source terms for vibrational and electron-electronic energy, as seen in Eqs. (10b) and (10c), are constructed to reflect these interactions. The governing equations of multi-T models with integrated energy mode can be obtained by summing each energy conservation equation as described in Sec. II A

The rotational energy relaxation source terms are as follows:

$$S_{trans-rot} = \sum_{s=mol}\left(\rho_s \frac{e_{rot,s}^{T_{trans}} - e_{rot,s}^{T_{rot}}}{\tau_s^{t-r}}\right), \tag{11a}$$

$$\tau_s^{t-r} = \frac{\sum_{r=mol} X_r}{\sum_{r=mol} \frac{X_r}{\tau_{sr}^{t-r}}}, \quad \tau_{sr}^{t-r} = \frac{1}{p}\exp\left(a_r T_{trans}^{-\frac{1}{3}} + b_r\right), \tag{11b}$$

$$S_{rot-e} = \sum_{s=mol}\left(g_{rot,s} 3\rho_e R(T_{rot} - T_e)\left(\frac{\nu_{e,s}}{M_s}\right)\right), \tag{12}$$

$$S_{vib-rot} = \sum_{s=mol}\left(\rho_s \frac{e_{vib,s}^{T_{vib}} - e_{vib,s}^{T_{rot}}}{\tau_s^{v-r}} f_{vib-rot,s}\right). \tag{13}$$

Equation (11a) is the *trans* – *rot* relaxation term in the form of the Landau–Teller equation.[34] $e_{rot,s}^{T_{trans}}$, $e_{rot,s}^{T_{rot}}$ are the rotational energies of species s calculated by the translational and rotational temperature. τ_s^{t-r} is the *trans* – *rot* relaxation time of species s from Jo *et al.*[26] The coefficients of the *trans* – *rot* relaxation time model applied to the multi-T models is summarized in Table I. Equation (12) is the energy relaxation term between a heavy particles and electrons by Nishida and Matsumotob.[20] $\nu_{e,s}$ is the effective collision frequency of $S_{trans-e}$, which will be discussed in Eq. (18). $g_{rot,s}$ is the ratio of $S_{trans-e}$ to S_{rot-e}, with values of 10 for nitrogen and oxygen, and 100 for nitric oxide. $g_{rot,s}$ of an ion is equal to the its value for the corresponding molecule. Equation (13) represents the *vib* – *rot* relaxation term in the form of the Landau–Teller equation.[34] The vibrational energies of species s, denoted by $e_{vib,s}^{T_{vib}}$ and $e_{vib,s}^{T_{rot}}$, are calculated by the vibrational and rotational temperatures, respectively. The vibrational-rotational relaxation time τ_s^{v-r} is obtained by omitting the high-temperature correction term from τ_s^{t-v}, which will be discussed in Eq. (15).[32] $f_{vib-rot,s}$ represents the fractional contribution of the *vib* – *rot* to the total energy transfer developed by Kim and Boyd.[31] The rotational energy chemical source term is as follows:

$$S_{chem-rot} = \sum_{s=mol}\left((\dot{\omega}_{exc,s} + \dot{\omega}_{aio,s} + \dot{\omega}_{cex,s})e_{rot,s} + Mol\right), \tag{14a}$$

TABLE I. Coefficient of translational-rotational relaxation time.

Type of collision[a]	M	a_r	b_r	Source
$N_2 + M$	N_2	−44.3800	−14.9915	Jo *et al.*[26]
	O_2	−42.6384	−15.6476	
	NO	−42.6384	−15.6942	
	N	−39.7430	−16.8270	
	O	−42.6384	−15.5886	
$O_2 + M$	N_2	−43.8486	−15.7097	
	O_2	−38.5430	−15.5170	
	NO	−43.8486	−15.6989	
	N	−43.8486	−15.7801	
	O	−36.4220	−14.4070	
$NO + M$	N_2	−46.0090	−16.6808	
	O_2	−46.0090	−16.6233	
	NO	−46.0090	−16.6704	
	N	−46.0090	−16.7482	
	O	−46.0090	−16.5711	

[a]The coefficient of collision with a charged species is identical to that of the corresponding neutral species.

$$Mol = (\dot{\omega}_{dis_F,s} + \dot{\omega}_{eid_F,s})\beta_{rot,s}E_s^D + (\dot{\omega}_{dis_R,s} + \dot{\omega}_{eid_R,s})e_{rot,s}, \tag{14b}$$

$$\beta_{rot,s} = \exp(A_{r_1}/T_{trans} + A_{r_2} + A_{r_3}\ln(T_{trans}) + A_{r_4}T_{trans} + A_{r_5}T_{trans}^2). \tag{14c}$$

The subscript of species production rates denotes the classification of the chemical reaction types. The subscript *exc* denotes exchange reaction, *aio* denotes associative ionization reaction, and *cex* denotes the charge exchange reaction. The term *Mol* in Eq. (14b) describes the rotational energy loss due to dissociation (*dis*) and electron impact dissociation reaction (*eid*) chemical reactions. The subscript F and R denote the forward and reverse reactions, respectively. In computing the rotational energy loss involving the dissociation reaction, Kim and Jo[33] proposed a preferential model for the forward reaction and a non-preferential model for the backward reaction. The rotational energy removal ratio $\beta_{rot,s}$ by Jo *et al.*[26] is used in Eq. (14b). The fitted coefficients of Eq. (14c) are listed in Table II. The parameter E_s^D is the dissociation energy of the species molecule.

The vibrational energy relaxation source terms are as follows:

$$S_{trans-vib} = \sum_{s=mol}\left(\rho_s \frac{e_{vib,s}^{T_{trans}} - e_{vib,s}^{T_{vib}}}{\tau_s^{t-v}}(1 - f_{vib-rot,s})\right), \tag{15}$$

$$S_{e-vib} = \sum_{s=N_2,O_2,NO}\left(\rho_s \frac{e_{vib,s}^{T_e} - e_{vib,s}^{T_{vib}}}{\tau_s^{e-v}}\right). \tag{16}$$

Equation (15) represents the *trans* – *vib* relaxation in the form of the Landau–Teller equation,[34] where $e_{vib,s}^{T_{trans}}$, $e_{vib,s}^{T_{vib}}$ represent the vibrational energy of species s calculated by the translational and vibrational temperature, respectively. τ_s^{t-v} is the *trans* – *vib* relaxation time from Kim and Jo.[33] Equation (16) represents the *e* – *vib* relaxation in the form of the Landau–Teller equation.[34] $e_{vib,s}^{T_e}$, $e_{vib,s}^{T_{vib}}$ are the vibrational energies of species s calculated by electron and vibrational temperature, respectively. τ_s^{e-v} is the *e* – *vib* relaxation time, proposed by Lee[16] for nitrogen and Laporta *et al.*[18,19] for oxygen and nitric oxide. The *e* – *vib* relaxation time models applied to the multi-T models are summarized in Table III. The vibrational energy chemical source term is as follows:

$$S_{chem-vib} = \sum_{s=mol}\left((\dot{\omega}_{exc,s} + \dot{\omega}_{aio,s} + \dot{\omega}_{cex,s})e_{vib,s} + Mol\right), \tag{17a}$$

$$Mol = (\dot{\omega}_{dis_F,s} + \dot{\omega}_{eid_F,s})\beta_{vib,s}E_s^D + (\dot{\omega}_{dis_R,s} + \dot{\omega}_{eid_R,s})e_{vib,s}, \tag{17b}$$

$$\beta_{vib,s} = \exp(A_{v_1}/T_{trans} + A_{v_2} + A_{v_3}\ln(T_{trans}) + A_{v_4}T_{trans} + A_{v_5}T_{trans}^2). \tag{17c}$$

Equation (17a) is the vibrational energy loss due to chemical reaction, similar to Eq. (14a). The term *Mol* in Eq. (17b) is calculated in a similar way as in Eq. (14b). $\beta_{vib,s}$ of Eq. (17b) represents the vibrational energy removal ratio by Kim and Jo,[33] and the fitted coefficients are listed in Table IV.

Electron-electronic energy relaxation source terms are

$$S_{trans-e} = \sum_{s\neq e}\left(3\rho_e R(T_{trans} - T_e)\left(\frac{\nu_{e,s}}{M_s}\right)\right). \tag{18}$$

Equation (18) represents the energy relaxation of heavy particles and electrons due to elastic collisions introduced by Appleton and Bray.[14] $\nu_{e,s}$ is the effective collision frequency. The effective collision cross

TABLE II. Fitted coefficient of rotational energy removal ratio.

Species[a]	A_{r_1}	A_{r_2}	A_{r_3}	A_{r_4}	A_{r_5}
N_2[b]	1.360×10^3	-1.308×10^1	1.215×10^0	-4.915×10^{-5}	3.113×10^{-10}
O_2[c]	-1.712×10^2	-8.118×10^0	7.386×10^{-1}	-4.898×10^{-5}	4.383×10^{-10}
NO	0.000×10^0	-1.609×10^0	0.000×10^0	0.000×10^0	0.000×10^0

[a]The coefficient of removal ratio of charged species is identical to that of the corresponding neutral species.
[b]The coefficients are applicable within the temperature range of 1000–30 000 K.
[c]The coefficients are applicable within the temperature range of 1000–30 000 K.

TABLE III. Models for relaxation time of electron impact vibrational excitation process.

Species	τ_s^{e-v}	Temperature range (K)	Source
N_2	$\log(p_e\tau_{N_2}^{e-v}) = \alpha(\log(T_e))^2 + \beta\log(T_e) + \gamma$	$1000 < T_e < 50\,000$	Lee[a]
O_2	$p_e\tau_{O_2}^{e-v} = \sum_{n=0}^{4}\sum_{m=0}^{2} c_{n,m} T_e^n T_{vib}^m$	$2000 < T_e < 20\,000, 0 < T_{vib} < 20\,000$	Laporta *et al.*[b]
NO	$n_e\tau_{NO}^{e-v} = a_0 + a_1 T_e + a_2 T_e^2 + b/\ln T_e$	$200 < T_e < 50\,000, 100 < T_{vib} < 20\,000$	Laporta *et al.*[c]

[a]$\alpha = 3.91$, $\beta = -30.36$, $\gamma = 48.9$ for $1000 \leq T_e < 7000$. $\alpha = 1.30$, $\beta = -9.09$, $\gamma = 5.58$ for $7000 \leq T_e < 50\,000$.
[b] The detailed value of the coefficient $c_{n,m}$ can be found in Laporta *et al.*[18]
[c] The detailed value of the coefficient a_0, a_1, a_2, b can be found in Laporta *et al.*[19]

TABLE IV. Fitted coefficient of vibrational energy removal ratio.

Species[a]	A_{v_1}	A_{v_2}	A_{v_3}	A_{v_4}	A_{v_5}
N_2[b]	-4.724×10^2	1.614×10^0	-2.821×10^{-1}	-4.826×10^{-6}	1.193×10^{-10}
O_2[c]	1.044×10^0	-7.883×10^{-1}	-4.532×10^{-3}	-5.011×10^{-5}	9.229×10^{-10}
NO	0.000×10^0	-1.204×10^0	0.000×10^0	0.000×10^0	0.000×10^0

[a]The coefficient of removal ratio of charged species is identical to that of the corresponding neutral species.
[b]The coefficients are applicable within the temperature range of 1000–30 000 K.
[c]The coefficients are applicable within the temperature range of 1000–30 000 K.

section from Gnoffo[35] is used for electron-neutral energy transfer. For electron-ion energy transfer, Scalabrin's effective collision cross section is applied.[36] Electron-electronic energy chemical source term is as follows:

$$S_{chem-ee} = \sum_{s\neq e}((\dot{\omega}_{dis,s} + \dot{\omega}_{exc,s} + \dot{\omega}_{aio,s} + \dot{\omega}_{cex,s} + \dot{\omega}_{eid,s})e_{el,s}) + \sum_{s=atom}(Atom) + (\dot{\omega}_{aio,e} + \dot{\omega}_{eii,e})e_{e,e}, \tag{19a}$$

$$Atom = \begin{cases} (\dot{\omega}_{eii_F,s})\alpha_s E_s^I + (\dot{\omega}_{eii_R,s})e_{el,s} & \text{for Neutral,} \\ (\dot{\omega}_{eii_F,s})e_{el,s} + (\dot{\omega}_{eii_R,s})e_{el,s} & \text{for Charged,} \end{cases} \tag{19b}$$

$$\alpha_s = 0.3. \tag{19c}$$

Equation (19a) represents the electron-electronic energy loss due to a chemical reaction. *Atom* in Eq. (19b) is the electronic energy loss of a neutral, charged atom due to *eii* (electron impact ionization). Unlike charged atoms, neutral atoms use the preferential model for forward reactions and the non-preferential model for reverse reactions. In Eq. (19b), α_s is the electronic energy removal ratio by Kim and Jo,[33] and E_s^I is the ionization potential.

C. Chemical kinetics of multi-temperature models

To accurately describe the species production rate, the modified Arrhenius parameter by Kim and Jo,[33] which considers various chemical reactions channels ($dis, exc, aio, cex, eid, eii$), is used. Production rate by chemical reaction $\dot{\omega}_s$ in Eq. (1) is described as

$$\dot{\omega}_s = M_s \sum_{k=1}^{NRT}(\beta_{k,s} - \alpha_{k,s})[R_{f,k} - R_{b,k}], \tag{20}$$

where M_s is the molar mass, and $\alpha_{k,s}$ and $\beta_{k,s}$ are the stoichiometric coefficients of the reactants and products in the kth reaction. Factors of 10^3, 10^{-3} of the forward and reverse reaction rates, $R_{f,k}$, $R_{b,k}$, are used to convert CGS units of reaction parameters to MKS units,

$$R_{f,k} = 10^3 \times k_{f,k} \prod_s \left(10^{-3} \times \frac{\rho_s}{M_s}\right)^{\alpha_{k,s}}, \quad R_{b,k} = 10^3 \times k_{b,k} \prod_s \left(10^{-3} \times \frac{\rho_s}{M_s}\right)^{\beta_{k,s}}, \tag{21}$$

where $k_{f,k}$ and $k_{b,k}$ represent the forward and reverse rate coefficients for the kth reaction, respectively.

$$k_{f,k} = C_{f,k} T_f' \eta_k e^{-\frac{\theta_k}{T_f'}}, \quad k_{b,k} = \frac{k_{f,k}(T_b')}{K_e(T_b')}, \tag{22}$$

where $C_{f,k}$, η_k, and θ_k are the Arrhenius parameters of the kth reaction. T_f', T_b' are the forward and reverse reaction control temperatures, respectively. Reverse reaction control temperature can be obtained similarly,

$$T_f' = \frac{1}{2}\left[(T_f + T_{min}) + \sqrt{(T_f - T_{min})^2 + \epsilon^2}\right], \quad \epsilon = 0.1 \times T_{min}. \tag{23}$$

It should be noted that the equilibrium constant K_e in Eq. (22) is calculated using the reverse reaction control temperature. Equilibrium constant parameters of the modified 2-T of Kim and Jo[33] are applied

$$\ln(K_e(T)) = \frac{A_1}{Z} + A_2 + A_3 \ln(Z) + A_4 Z + A_5 Z^2, \quad Z = \frac{10\,000}{T}. \tag{24}$$

D. Thermodynamic property of multi-temperature models

The pressure of a mixture is the sum of the partial pressures of all species. The partial pressure can be obtained from the equation of state using the translational temperature of each species. The electron temperature refers to the translational temperature of the free electron. Based on this, the flow pressure can be obtained as

$$p = p_h + p_e = \sum_{s\neq e}^{all} \rho_s \frac{R}{M_s} T_{trans} + \rho_e \frac{R}{M_e} T_e, \tag{25}$$

where R is the universal gas constant. The total energy of the mixture is

$$\rho e = E = E_{trans} + E_{rot} + E_{vib} + E_e + E_{el} + E_{HoF} + E_{KE}. \tag{26}$$

In Eq. (26), E_{trans}, E_{rot}, E_{vib}, E_e, E_{el}, E_{HoF}, and E_{KE} represent the translational, rotational, vibrational, electron, electronic, heat of formation, and kinetic energies per unit volume, respectively. Detailed descriptions for each energy are as follows:

$$E_{trans} = \rho e_{trans} = \sum_{s\neq e}^{all} \rho_s \left(\frac{3}{2}\frac{R}{M_s}\right) T_{trans}, \tag{27a}$$

$$E_{rot} = \rho e_{rot} = \sum_{s=mol} \rho_s \left(\frac{R}{M_s}\right) T_{rot}, \tag{27b}$$

$$E_{vib} = \rho e_{vib} = \sum_{s=mol} \rho_s \left(\frac{R}{M_s}\right)\left(\frac{\theta_{vib,s}}{e^{\theta_{vib,s}/T_{vib}-1}}\right), \tag{27c}$$

$$E_e = \rho e_e = \rho_e \left(\frac{3}{2}\frac{R}{M_e}\right) T_e, \tag{27d}$$

$$E_{el} = \rho e_{el} = \sum_{s\neq e} \rho_s \left(\frac{R}{M_s}\right)\left(\frac{\sum_{i=1}^{\infty} g_{i,s}\theta_{el,i,s} e^{-\theta_{el,i,s}/T_{el}}}{\sum_{i=0}^{\infty} g_{i,s} e^{-\theta_{el,i,s}/T_{el}}}\right), \tag{27e}$$

$$E_{HoF} = \rho e_{HoF} = \sum_{s=1}^{all} \rho_s \frac{\Delta h_s^{\circ}}{M_s}, \tag{27f}$$

$$E_{KE} = \rho e_{KE} = \sum_{s=1}^{all} \rho_s \frac{1}{2}(u^2 + v^2). \tag{27g}$$

In Eq. (27d), $\theta_{vib,s}$ is the characteristic vibrational temperature. $g_{i,s}$ and $\theta_{el,i,s}$ in Eq. (27f) are the degeneracy of the ith level of electronic energy and the characteristic electronic temperature, which is obtained from Mutation++.[37] The total enthalpy of the mixture is

$$\rho h = \rho e + p. \tag{28}$$

The notation for electron-electronic energy and enthalpy are defined as

$$\rho e_{ee} = \rho e_e + \rho e_{el}, \rho h_{ee} = \rho e_{ee} + p_e. \tag{29}$$

E. Transport property of multi-temperature models

The transport properties of the multi-T models are obtained using Gupta's mixing rule.[38] The viscous stress tensor of Eqs. (2) and (3) is obtained as follows:

$$\tau_{xx} = \frac{4}{3}\mu\frac{\partial u}{\partial x}, \quad \tau_{yy} = \frac{4}{3}\mu\frac{\partial v}{\partial y}, \quad \tau_{xy} = \tau_{yx} = \mu\left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\right). \tag{30}$$

In Eq. (4), viscous diffusion in the x-direction is shown in Eq. (31). A similar formulation can also be applied for the viscous diffusion in the y-direction,

$$q_x = q_{trans,x} + q_{rot,x} + q_{vib,x} + q_{e,x} + q_{el,x}. \tag{31}$$

A detailed description of Eq. (31) is as follows:

$$q_{trans,x} = \sum_{s\neq e} J_{s,x} h_{trans,s} - \eta_{trans}\frac{\partial T_{trans}}{\partial x}, \tag{32a}$$

$$q_{rot,x} = \sum_{s=mol} J_{s,x} e_{rot,s} - \eta_{rot}\frac{\partial T_{rot}}{\partial x}, \tag{32b}$$

$$q_{vib,x} = \sum_{s=mol} J_{s,x} e_{vib,s} - \eta_{vib}\frac{\partial T_{vib}}{\partial x}, \tag{32c}$$

$$q_{e,x} = J_{e,x} h_{e,s} - \eta_e \frac{\partial T_e}{\partial x}, \tag{32d}$$

$$q_{el,x} = \sum_{s\neq e} J_{s,x} e_{el,s} - \eta_{el}\frac{\partial T_{el}}{\partial x}, \tag{32e}$$

where μ, η_{trans}, η_{rot}, η_{vib}, η_e, and η_{el} are the viscosity of the mixture and thermal conductivity of each energy mode. A detailed description of the transport properties is as follows:

$$\mu = \sum_{s=all} \frac{\left(\frac{M_s}{N_A}\right) X_s}{\sum_{r=all} X_r \Delta_{sr}^{(2)}}, \tag{33a}$$

$$\eta_{trans} = \frac{15}{4} k_B \sum_{s\neq e} \frac{X_s}{\sum_{r=all} a_{sr} X_r \Delta_{sr}^{(2)}}, \tag{33b}$$

$$\eta_{rot} = k_B \sum_{s=mol} \frac{X_s}{\sum_{r=all} X_r \Delta_{sr}^{(1)}}, \tag{33c}$$

$$\eta_{vib} = k_B \sum_{s=mol} \left(\frac{c_{v,vib,s}}{R}\right) \frac{X_s}{\sum_{r=all} X_r \Delta_{sr}^{(1)}}, \tag{33d}$$

$$\eta_e = \frac{15}{4} k_B \frac{X_s}{\sum_{r=all} 1.45 X_r \Delta_{er}^{(2)}}, \tag{33e}$$

$$\eta_{el} = k_B \sum_{s \neq e} \left(\frac{c_{v,el,s}}{R}\right) \frac{X_s}{\sum_{r=all} X_r \Delta_{sr}^{(1)}}, \tag{33f}$$

where k_B, N_A, X_s, and $c_{v,s}$ are the Boltzmann constant, the Avogadro number, the mole fraction, and the specific heat at constant volume of species s, respectively. $\Delta_{sr}^{(1)}$ and $\Delta_{sr}^{(2)}$ represent the Gupta's diffusion and viscous modified collision integral, respectively. The modified collision integral is calculated using the translational temperature, or the electron temperature if the collision involves electron.

III. RESULTS AND DISCUSSION

Numerical simulations of multi-T models (2-T, 3-T, 4-T) were conducted to compare the effects of nonequilibrium due to the separation of energy modes under various ground facility and flight test conditions. Experimental data of Radio Attenuation Measurements (RAM-C II)[39,40] were used to validate the electron number density of the multi-T models. Electre model,[41] Large Energy National Shock tunnels (LENS-XX) cylinder,[42] and Flight Investigation Reentry Environment (FIRE-II)[43] simulations were performed to compare the effects of electron and surface heat flux downstream of the bow shock. The LENS-XX double cone and the hollow cylinder-flare model[44] were simulated to verify the effect of electron and surface heat flux due to shock–shock interactions in multi-T models in the shock wave boundary layer interactions (SWBLI) phenomenon. The incoming freestream air was set to be 76% nitrogen and 24% oxygen in terms of mass fraction.

The CFD solver SHOCK2D,[33] which uses the finite volume method, was used for numerical simulation. The modified Steger–Warming method[45] was applied for calculating the inviscid flux, and the central difference method was used for calculating the viscous flux. The MUSCL[46] scheme with MINMOD[47] limiter was applied to obtain second order accuracy in space for accurate shock capturing, and the fully implicit backward Euler time integration scheme was applied. For all analysis cases, the height of the first layer at the wall was set to 1×10^{-6} m, and the converged grid, with grid independence test based on the 2-T model, was used.

A. RAM-C II

A Radio Attenuation Measurement-C II[39,40] flight test was conducted to analyze and mitigate the blackout phenomenon in a hypersonic vehicle. Blackout phenomena can be predicted by accurate flow and chemical reactions, and together with OREX,[48] the RAM-C II flight test is widely cited to validate the electron number density of CFD solvers. Numerical simulations were performed to validate the electron number density of multi-T models. In addition to the windward region, which can be directly compared to flight test data, the simulation domain was extended to the wake region expanding by the base geometry to determine the effect of temperature and electron number density. In the flight test, reflectometers and electrostatic probes were used to measure the electron number density. The reflectometer transmits electromagnetic waves in the normal direction of the surface and measures the intensity of the electromagnetic waves reflected by the electron layer to determine the electron number density corresponding to a specific frequency. The electrostatic probe can measure electron number density with an accuracy of up to 10^{19} 1/m^3. However, since the probe is installed at the rear of the vehicle, both the flow field and chemical reactions should be captured precisely in the numerical simulation to have a good comparison with the experimental data. RAM-C II consists of a sphere with a radius of 0.1524 m and a 9° cone with a length of 1.295 m. During the flight test, a beryllium nose cap was used up to an altitude of 56.39 km to minimize the contamination of the electron number density by the ablation. Numerical simulations were performed at altitudes of 61 and 71 km considering the flow condition, the magnitude of uncertainty, and the effect of contamination. Flight conditions of RAM-C II are listed in Table V. The wall boundary conditions were assumed to be non-catalytic for neutral species and fully catalytic for charged species.

Figure 1 shows reflectometer results from multi-T models and other numerical studies[25,33,36,49–51] with flight test data[39,40] for 61 and 71 km altitude conditions. The error bars in the reflectometer data indicate the critical electron number density corresponding to the boundaries of the L, S, X, and Ka band frequencies.[51] Both the 61 and 71 km conditions show high electron number density near the nose as a result of the strong bow shock. It then expands and decreases rapidly at the shoulder, and both multi-T models predicted the electron number density within the error bars. The symbols in the reflectometer data represent interpolated values; the further outside the error bar, the closer to the true value. The results of the multi-T models are in close agreement with the flight data at the first reflectometer of the 61 km condition and the first, third, and fourth reflectometers of the 71 km condition. The discrepancy in reflectometer results among the multi-T models is negligible, confirming that the separation of energy modes does not significantly affect the maximum electron number density normal to the wall.

Figure 2 compares the stagnation line temperature distributions for the 61 and 71 km altitude conditions for the multi-T models. The maximum vibrational temperature of the 3-T model increases slightly more than 2-T model due to the separation of electron-electronic energy. The difference between the vibrational and electron-electronic temperatures at 3-T is significant right after the shock wave. While the electron-electronic temperature increases upstream of the shock wave due to the influence of electron diffusion, the vibrational temperature shows an increase after the translational temperature increases. The difference in translational-rotational temperature between the 3-T and 2-T models is negligible. The 4-T model, which separates the rotational energy, shows an increase in the thermochemical nonequilibrium

TABLE V. Flight condition of RAM-C II experiment.

Altitude (km)	ρ_∞ (kg/m^3)	T_∞ (K)	T_{wall} (K)	U (m/s)
61	2.54×10^{-4}	243	1200	7650
71	6.42×10^{-5}	215	1200	7660

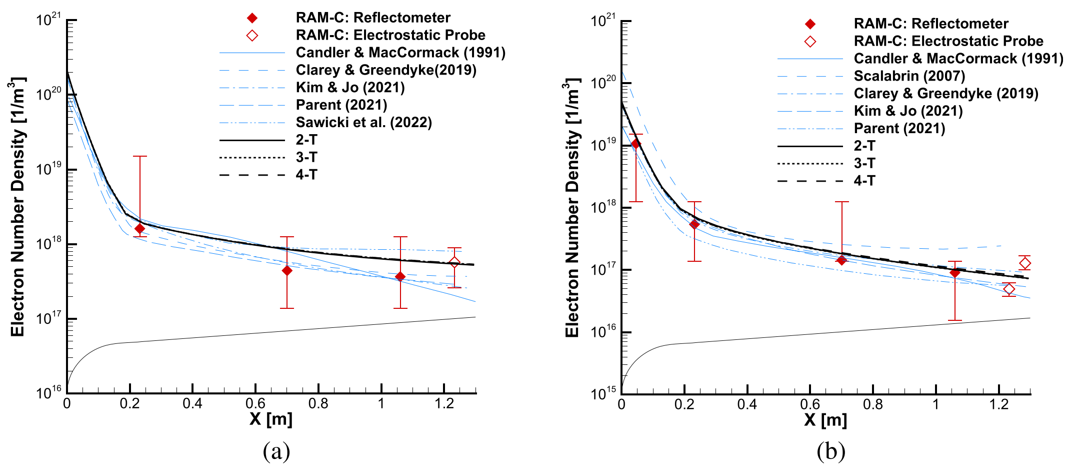


FIG. 1. Comparison of maximum electron number density normal to the wall of the RAM-C II flight test conditions: (a) Maximum electron number density of multi-T models at 61 km altitude with flight tests and other studies[25,33,49–51] and (b) maximum electron number density of multi-T models at 71 km altitude with flight tests and other studies.[25,33,36,49,50]

regime of the stagnation line due to the increase in shock standoff distance. The rotational temperature of the 4-T model has a similar temperature distribution to the vibrational temperature due to the increased relaxation time caused by the high translational temperature. Furthermore, the vibrational temperature is slightly increased due to the influence of the vibrational-rotational relaxation transfer. However, the maximum value of electron-electronic temperature and its distribution in the equilibrium regime are comparable to those observed in the 2-T and 3-T models. The multi-T models showed increased model-to-model differences due to the increase in thermochemical nonequilibrium effect at the 71 km altitude condition, where the density is lower. An analysis of the stagnation line temperatures shows that the discrepancies in the reflectometer results were relatively minor compared to the discrepancies in the temperature distributions between the multi-T models.

Figure 3 shows the production rate and number density of electrons along the stagnation line of the multi-T models at an altitude of 61 km. In the RAM-C II condition, most electrons at the stagnation line are produced by the *aio* channel, specifically by the $N + N = N_2^+ + e^-$ reaction. This differs from the results of other studies[22,49,51–53] that show that the majority of electrons are generated by the $N + O = NO^+ + e^-$ reaction. The modified 2-T parameter of Kim and Jo,[33] which only alters the parameters of the *dis* channel of the 2-T parameter of Park and coworkers,[54,55] shows good agreement with the flight test data for electron number density, despite changes in the main ionization reaction. The difference between the 2-T and 3-T models is significant at

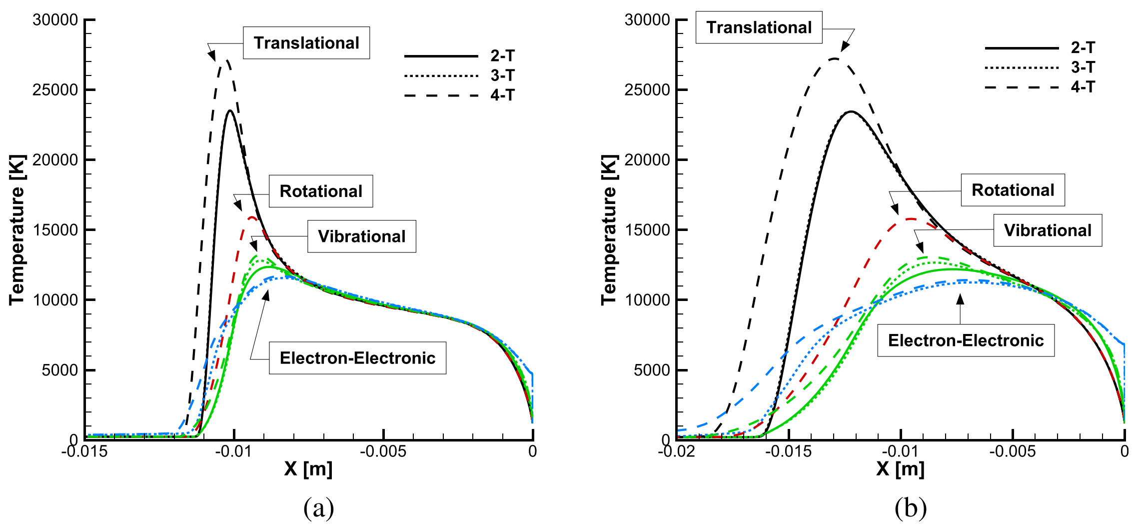
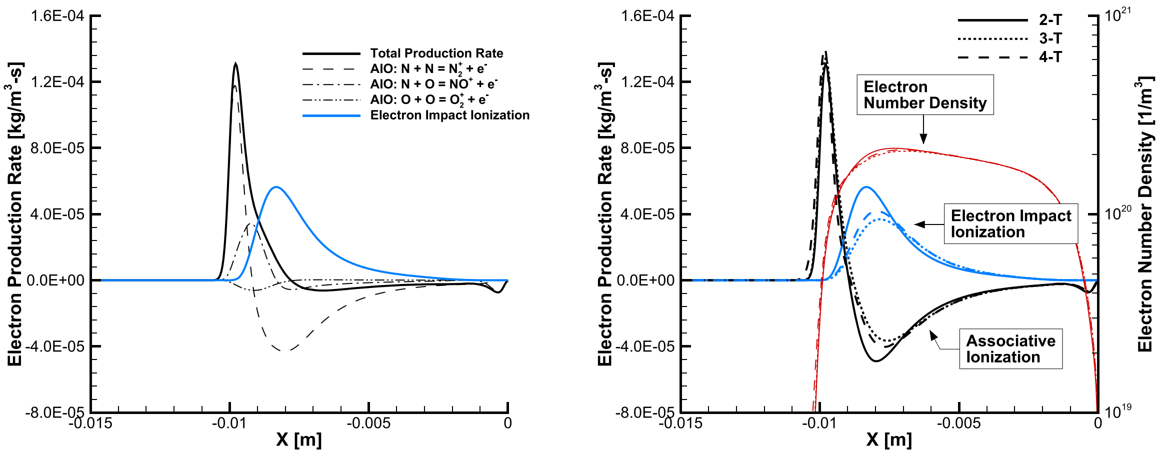


FIG. 2. Comparison of temperature distribution along the stagnation line of the multi-T models of the RAM-C II flight test conditions: (a) 61 km altitude conditions and (b) 71 km altitude conditions.

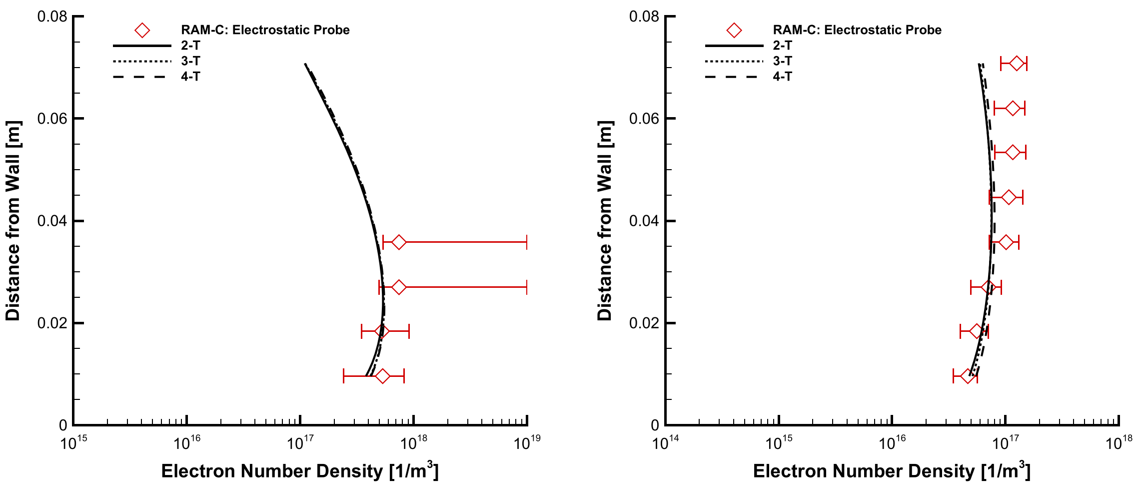


(a) (b)

FIG. 3. Electron production rate and number density along stagnation line of RAM-C II at 61 km: (a) Electron production rate of each reaction channel at 2-T model and (b) comparison of electron production rates between the multi-T models.

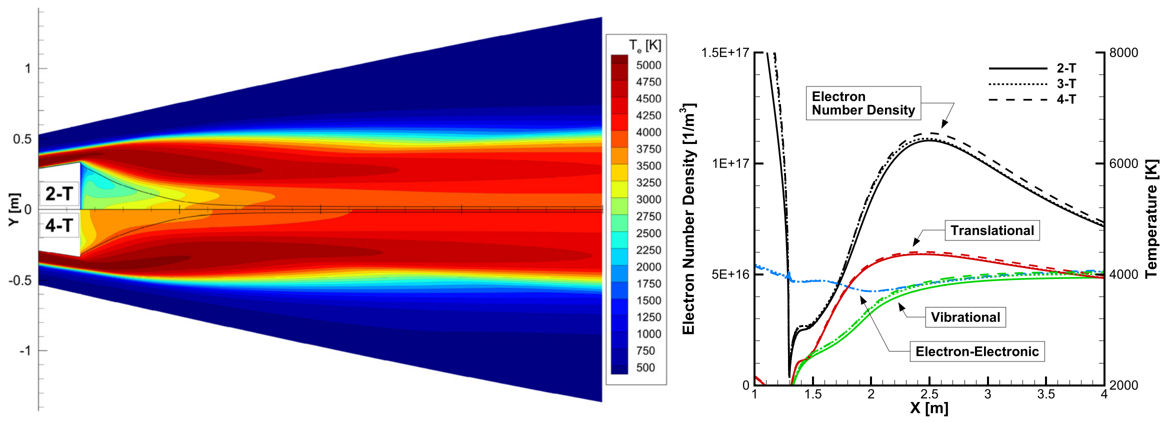
$x = -0.009$ m. At the stagnation line, electrons are mostly produced by the *aio* reaction right after the shock wave, and then the *aio*, *eii* reaction reaches equilibrium in the production and destruction of electrons. Chemical reactions involving electrons use the electron temperature to calculate the reaction rate. In the *aio* reverse reaction and *eii* forward and reverse reaction, the electron temperature was used to calculate the reaction rate. The discrepancy in electron temperature applied to the multi-T models resulted in a discrepancy in the electron production rate. However, in the regime where the effect of electron temperature dominates, the net production rate is so small that its impact on electron number density is negligible. The 4-T model produced electrons at an earlier location than the other models due to the increased shock standoff distance. However, the distribution of electrons in the regime where the net production rate decreased was similar to that of the 3-T model, resulting in a small change in the maximum electron number density.

Figure 4 compares the electron number density at the electrostatic probe of the multi-T models at 61 and 71 km altitude conditions with experimental data. In both flow conditions, the multi-T models show electron number densities within error, with differences between models up to 13.8% near the wall. As mentioned earlier, the comparison with the electrostatic probe data confirms that the multi-T models accurately predict the flow field and chemical reaction accurately after the strong shock. The discrepancy between the multi-T models results



(a) (b)

FIG. 4. Comparison of electron number density at the electrostatic probe of multi-T models with experimental data from the RAM-C II flight test: (a) 61 km altitude conditions and (b) 71 km altitude conditions.

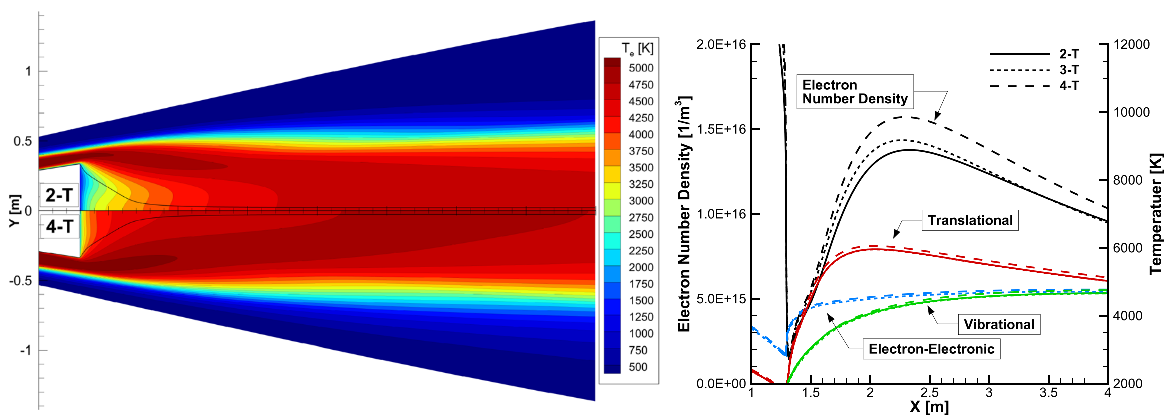


(a) (b)

FIG. 5. Comparison of temperature and electron distribution in wake region at 61 km altitude of RAM-C II flight test conditions: (a) Electron temperature distribution of 2-T and 4-T and (b) temperature and electron number density along streamline.

and the experimental data increases with increasing distance from the wall. This limitation arises from the axisymmetric solver used in this study due to the effect of oblique shock generated by the probe geometry.

Figures 5 and 6 show the effects of multi-T models in a wake region that expands rapidly after an electrostatic probe. Figure 5(a) compares the electron temperature of 2-T and 4-T models in the wake region at 61 km. The discrepancy in electron temperature between the two models is maximized in the recirculation region and gradually converges. The black streamline shown passes just above the recirculation region. The 4-T model changes the shock wave and flow structure, but the change in the y-axis direction for the selected streamline is within 0.1% of the nose radius. Figure 5(b) shows the temperature and electron number density of the streamline. The thermal nonequilibrium observed in the recirculation region occurred prior to the expansion. In contrast to the electron-electronic-vibrational and vibrational temperatures, which rapidly equilibrate with the translational temperature, the electron-electronic temperature shows a relatively long relaxation time during expansion and recompression shocks, consistent with the results of Clarey and Greendyke.[49] Both the 3-T and 4-T models predict higher electron density in the wake region compared to the 2-T model. In the 3-T model, the density decreases after reaching the maximum value caused by the recompression shock and converges to that of the 2-T model. However, the 4-T model does not converge after reaching its maximum value, which is higher than those of the 2-T and 3-T models, and this difference persists downstream. The 71 km condition shows significant thermochemical nonequilibrium in the wake region as well as the windward region due to the lower density than the 61 km condition. Figure 6(a) compares the electron temperatures in the wake region for the 2-T and 4-T models. Changes in the recirculation region are similar to the 61 km results, but thermal nonequilibrium persists farther downstream. Figure 6(b) compares the temperature and electron number



(a) (b)

FIG. 6. Comparison of temperature and electron distribution in wake region at 71 km altitude of RAM-C II flight test conditions: (a) Electron temperature distribution of 2-T and 4-T and (b) temperature and electron number density along streamline.

TABLE VI. Flight condition of Electre experiment.

Flight time (s)	Altitude (km)	ρ_∞ (kg/m^3)	T_∞ (K)	T_{wall} (K)	U (m/s)
293	53.3	6.963×10^{-4}	265	343	4230

density of the streamline with the same methodology as Fig. 5(b). The 71 km condition results in a greater temperature difference among the multi-T models compared to the 61 km condition, leading to a greater difference in electron number density.

B. Electre

Electre,[41] selected as the standard model for European Hypersonic Facilities, has been the subject of several subsequent studies analyzing the surface heat flux.[56–59] Numerical simulations were performed to analyze the effects of multi-T models on the surface heat flux and electron density under Electre flight condition. Electre consists of a sphere with a radius of 0.175 m and a 4.6° cone with a length of 2 m. The simulation was performed under flight test condition of 293 s, taking into account the published experimental data and the results of previous studies. The flow conditions of the Electre simulated

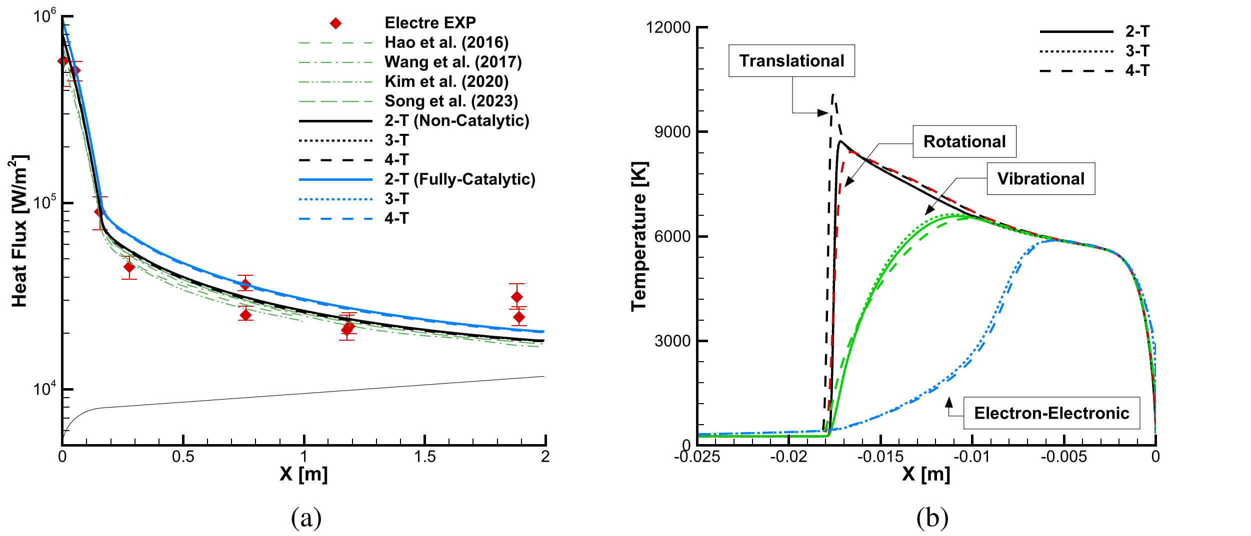


FIG. 7. Comparison of surface heat flux and stagnation line temperature distribution of the 293 s flight test condition of Electre: (a) Surface heat flux of multi-T models with flight test and other numerical studies[56–59] and (b) temperature distribution along the stagnation line of multi-T models.

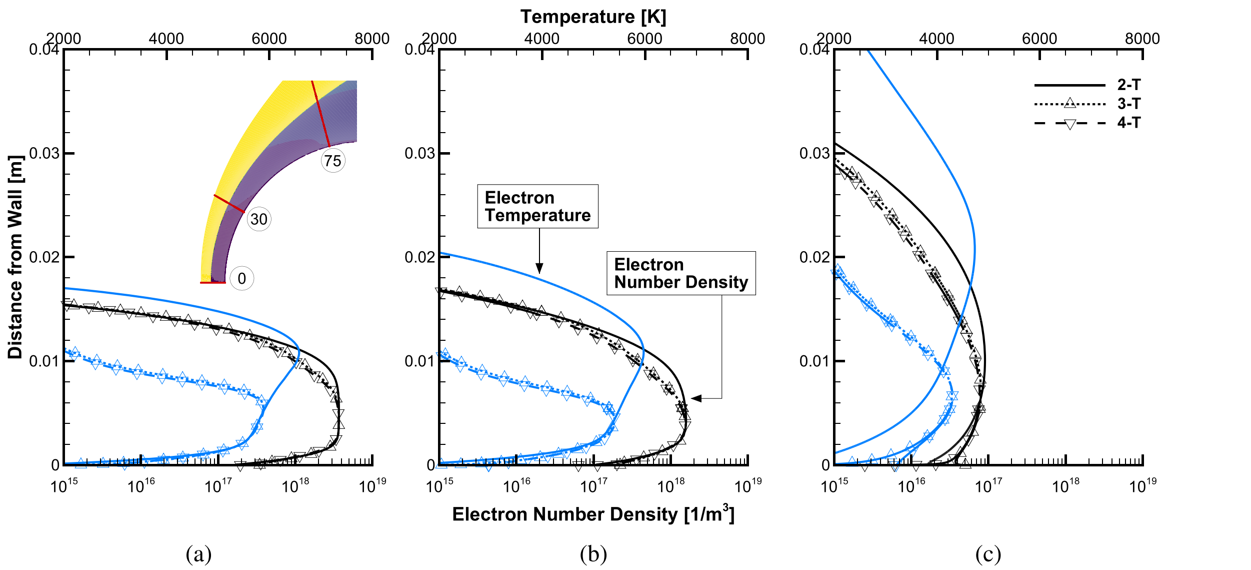


FIG. 8. Electron temperature and electron number density normal to the wall at (a) 0°, (b) 30°, and (c) 75° from the nose for the Electre flight test conditions with non-catalytic boundary condition.

TABLE VII. Free stream condition of the LENS-XX cylinder experiment.

Run	ρ_∞ (kg/m^3)	T_∞ (K)	T_{wall} (K)	U (m/s)	Test gas
73	0.68×10^{-3}	372	300	5623	Air
74	0.86×10^{-3}	377	300	5716	Air
75	0.47×10^{-3}	1070	300	7105	Air
77	0.90×10^{-3}	156	300	4502	Air

are listed in Table VI. Both non-catalytic and fully catalytic wall conditions are considered in the simulation.

Figure 7 shows the surface heat flux and temperature distribution along the stagnation line from multi-T models and other numerical studies[56–59] with experimental data[41] for 293 s condition. All of the multi-T models predicted the surface heat flux within the error, except for the experimental data at the end of the conical shape. As shown in Fig. 7(a), the discrepancies among the multi-T models are minor, with a maximum discrepancy of 2.1%. Figure 7(b) compares the stagnation line temperature distribution at 293 s of non-catalytic conditions for multi-T models. The flight condition of Electre is characterized by lower altitude and lower enthalpy in comparison to the previous RAM-C II. Consequently, the rotational temperatures reached equilibrium rapidly, whereas the electron-electronic temperature reached equilibrium slowly. The translational and rotational temperatures of the 4-T model are slightly higher than those of the 2-T and 3-T models. This is due to the additional *vib – rot* relaxation time considered in the 4-T model being longer than the *trans – vib* relaxation time used in the 2-T and 3-T models, resulting in a decrease in energy relaxation transfer.

Figure 8 compares the temperature and electron number density normal to the wall at the 0°, 30°, and 75° position from the nose

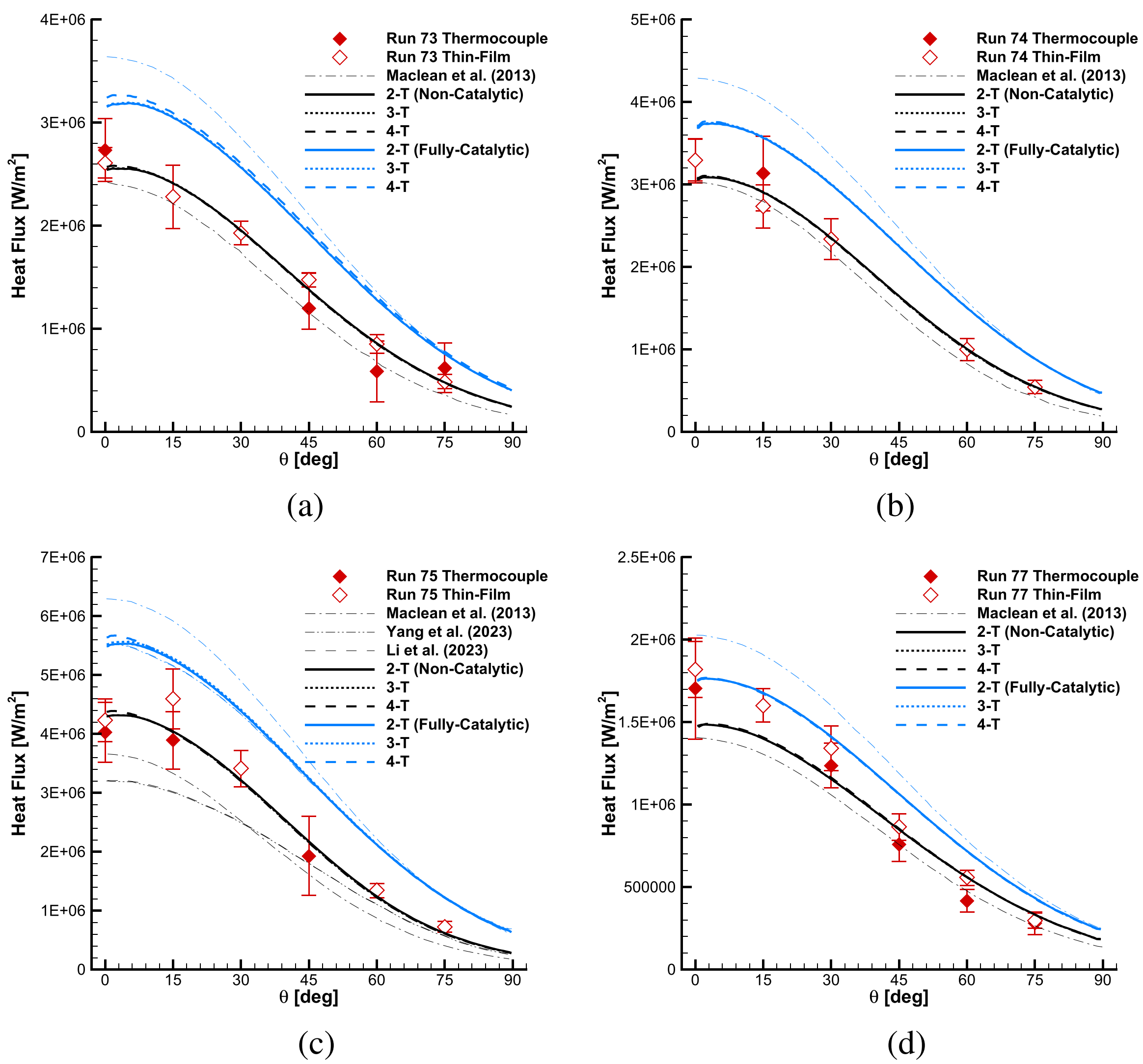


FIG. 9. Comparison of surface heat flux of multi-T models with LENS-XX cylinder test[42] and numerical simulation data:[42,61,62] (a) Run 73, (b) Run 74, (c) Run 75, and (d) Run 77.

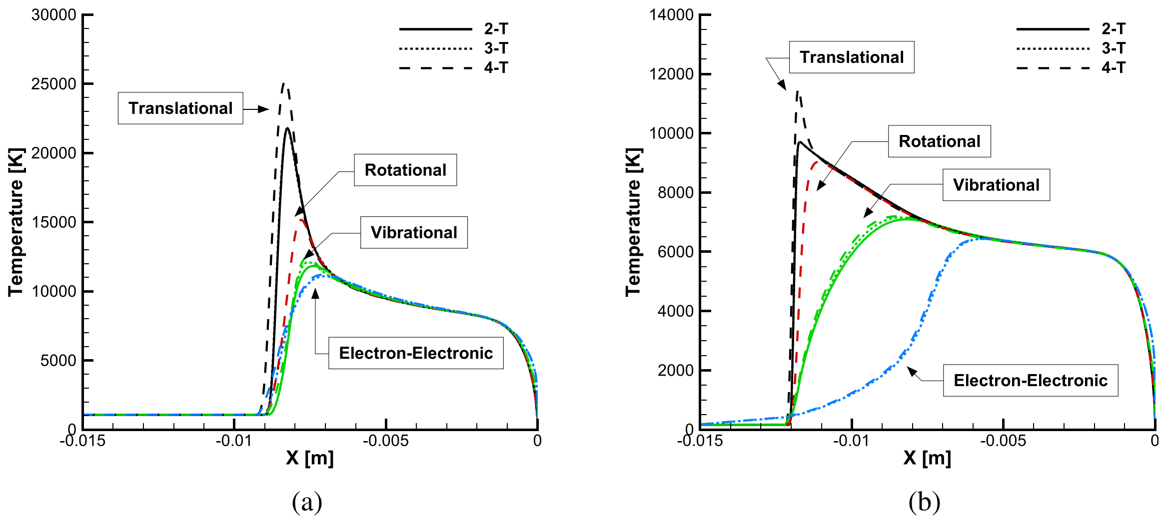


FIG. 10. Comparison of temperature distribution along stagnation line of multi-T models at LENS-XX cylinder free stream conditions: (a) Run 75 and (b) Run 77.

under non-catalytic conditions. At the 0°, 30° position, the flow is compressed after the shock wave and reaches equilibrium among the multi-T models near the wall. Although the difference in electron temperature leads to a difference in electron number density, the electron number density increases to a maximum at thermal equilibrium and the discrepancy of maximum electron number density among the multi-T models is negligible. The effect of rotational nonequilibrium is so small that the electron temperature and electron density in the 4-T model are almost identical to the 3-T model. At 75°, the flow expands to supersonic after the sonic line. The rapid expansion increases the difference in electron temperature among the multi-T models. The maximum electron number density decreases by 11.3% in the 3-T model, and both electron temperature and electron number density values do not converge near the wall. Even in the supersonic expansion region, the electron temperature and electron number density of the 4-T model are almost identical to those of the 3-T model.

C. LENS-XX: Cylinder

Another simulation considered is the cylinder model[42] conducted at the ground test facility of CUBRC's LENS-XX[60] expansion tunnel. Maclean *et al.*[42] compared experimental and numerical simulation

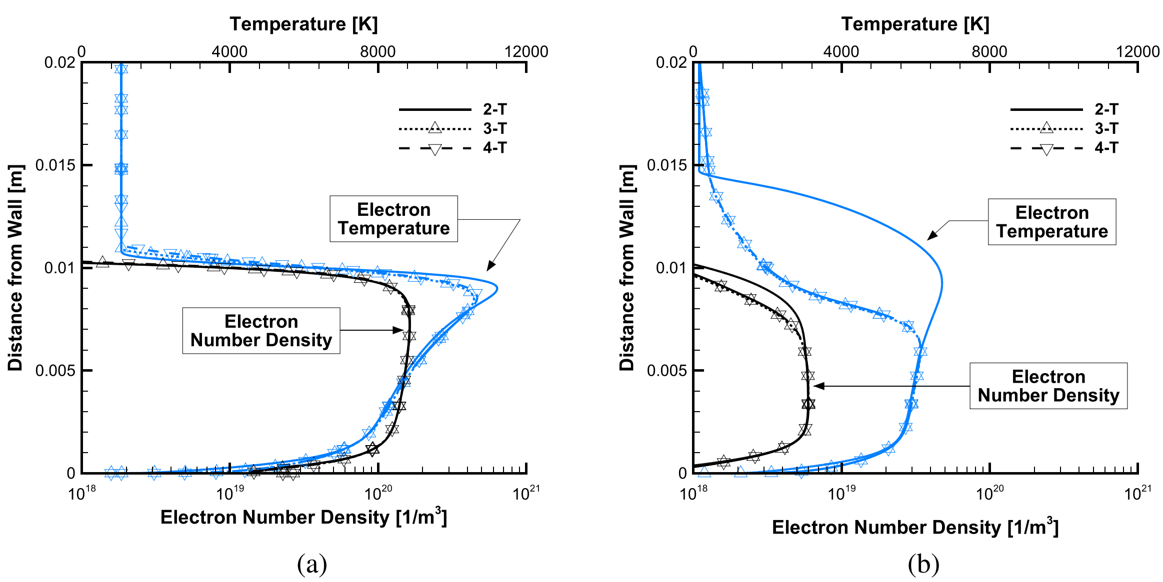


FIG. 11. Comparison of electron temperature and number density normal to the wall at 30° from the nose for LENS-XX cylinder free stream conditions: (a) Run 75 and (b) Run 77.

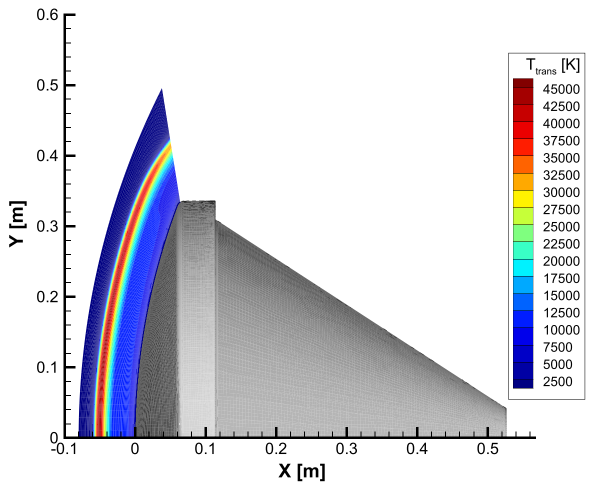


FIG. 12. FIRE-II geometry and translational temperature of 4-T model at 1634 s flight test condition.

results at high enthalpy conditions up to 26.07 MJ/kg using air as the test gas to determine the effect of the catalytic boundary condition. The effects of multi-T models on the surface heat flux and electrons downstream of a strong bow shock of a cylinder are investigated. The diameter of the cylinder is 8.89 cm, respectively. The flow conditions for each geometry are listed in Table VII. Both non-catalytic and fully catalytic wall boundary conditions were considered.

Figure 9 compares the heat flux distribution from multi-T models and numerical studies[42,61,62] for a cylinder with experimental data.[42,61,62] The cylinder model used air as the test gas for all flow conditions, and the effect of the catalytic wall is significant due to the dissociation of oxygen. All multi-T models predicted the surface heat flux within the error bars of the experimental data for both non-catalytic and fully catalytic conditions. With the exception of Run 77, the experimental data generally show good agreement with the non-catalytic results. Maclean *et al.* applied a super-catalytic boundary condition, explicitly setting the species mass fraction at the wall to ensure conservative predictions. This resulted in a higher surface heat flux compared to the fully catalytic condition. The increase in heat flux of the 4-T model near the stagnation point under Run 73 and 75 conditions is due to the misalignment of the shock and grid. The relatively low density in these free stream conditions increases the rate of change of the shock standoff distance, causing the shock position to move out of the aligned grid. Although the differences in heat flux between the multi-T models are small, there is little consistency between these results and those of other numerical simulations. This is because the heat flux of the cylinder model is strongly influenced by the boundary condition and thermochemical model.

TABLE VIII. Flight condition of FIRE-II experiment.

Flight time (s)	Altitude (km)	ρ_∞ (kg/m^3)	T_∞ (K)	T_{wall} (K)	U (m/s)
1634	76.42	3.72×10^{-5}	195	615	11 360
1636	71.02	8.57×10^{-5}	210	810	11 310
1637.5	67.05	1.47×10^{-4}	228	1030	11 250

Figure 10 compares the temperature distribution along the stagnation line of the multi-T models with non-catalytic wall boundary conditions for Run 75 and 77. These free stream conditions represent the highest and lowest enthalpy conditions of the cylinder model, respectively. Each flow condition is similar to RAM-C II 61 km and Electre, respectively, so the temperature distribution of the stagnation line and the characteristics of the multi-T models are very similar despite the different nose radius and spatial dimension. Figure 11 compares the electron temperature and number density normal to the wall of the multi-T models at the 30° position from the nose under the non-catalytic boundary condition of Run 75 and 77. The 30° position of the cylinder is the region where the flow is compressed and exhibits a similar temperature distribution to the stagnation line. The two conditions differ in total enthalpy by a factor of 2.6 and in degree of ionization by an order of 1.6, resulting in a significant difference in temperature and electron number density distribution. In Fig. 11(a), Run 75 quickly reaches thermochemical equilibrium with high temperature and pressure immediately after the shock wave, resulting only in a small difference in electron temperature and number density among the multi-T models. However, in Run 77, due to the relatively low temperature and degree of ionization, there is a substantial increase in the difference in electron temperature and number density among the multi-T models compared to Run 75.

D. FIRE-II

The FIRE-II flight test[43] was conducted to measure radiative and total heat flux at lunar return level velocities. FIRE-II flight data are widely cited for validating surface heat flux while accounting for radiation, and numerous subsequent studies[33,36,59,63–68] have been conducted. Numerical simulations were performed to investigate the effect of multi-T models on the electron and convective heat flux under FIRE-II flight conditions. The effect of radiation is assumed to be negligible. FIRE-II consists of a blunt nose and conical afterbody with a radius of curvature of 0.935 m. As shown in Fig. 12, only the forebody is set as the simulation domain. The numerical simulation was performed under the condition that the influence of radiation is negligible among the published experimental data. The flow conditions of FIRE-II analyzed are listed in Table VIII. For robust simulations, non-catalytic and super-catalytic wall boundary conditions are applied.

Figure 13 compares the surface heat flux of the multi-T models and other numerical studies[36,59,63,64,66,67] for the 1634, 1636, and 1637.5 s flight time conditions. Multi-T models predicted stagnation point heat flux comparable to other studies, while the 3-T and 4-T models predicted surface heat flux larger than the 2-T model. The three flight conditions have similar velocity and total enthalpy, but the free stream density increases with decreasing altitude, leading to higher heat flux. Near $y = 0.33$ m, the heat flux increases for all flow conditions due to the shoulder geometry of the forebody.

Figure 14 shows the comparison of stagnation point heat flux by each energy mode for multi-T models in non-catalytic conditions. The q_{trans} and q_{vib} for the 2-T model represent the translational-rotational and electron-electronic-vibrational heat flux, respectively, and the q_{trans} for the 3-T model represents the translational-rotational heat flux. The amount of change in heat flux increases as the altitude decreases. The increase in the 3-T and 4-T models is attributed to the

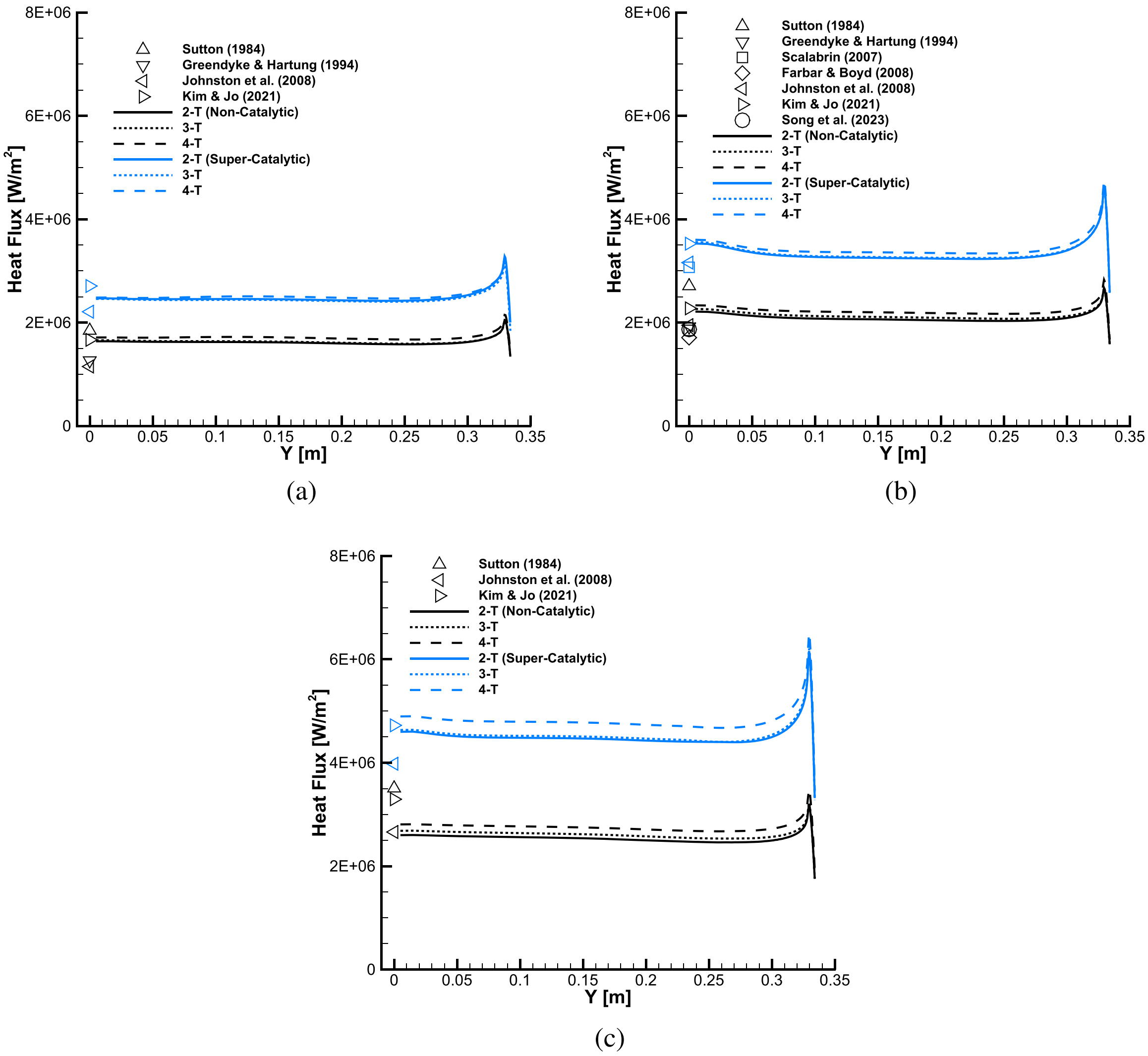


FIG. 13. Comparison of surface heat flux of multi-T models with numerical simulation data[33,36,59,63,64,66,67] at FIRE-II flight test conditions: (a) 1634, (b) 1636, and (c) 1637.5 s.

electron-electronic energy mode. The FIRE-II flight test condition has the largest degree of ionization among all of the flow conditions analyzed in this study, resulting in the second largest portion of heat flux resulting from electron-electronic energy. In the 3-T and 4-T models, the temperature distribution changes with separation and transfer of energy between energy modes. The electron-electronic temperature gradient near the wall increases significantly, resulting in an increase in heat flux in the separated energy model compared to the integrated energy model. The influence increases in higher density free stream conditions, with up to 8.0% in convective heat flux for the 4-T model.

Figure 15 compares the temperature distribution along the stagnation line for multi-T models with non-catalytic wall boundary conditions at 1634 and 1637.5 s. At the 1637.5 s condition, the degree of ionization and thermal nonequilibrium are significantly high, so the influence of free electrons on each temperature is notable. The vibrational and electron-electronic temperatures in the 3-T model are comparable to the electron-electronic-vibrational temperatures in the 2-T model after the shock wave and at the location of maximum temperature, respectively. In addition, a steep rise in the electron-electronic temperature upstream of the shock wave leads to an increase in the translational-rotational temperature. At 1637.5 s, the shock standoff distance and nonequilibrium regime decreased as a result of the high density. FIRE-II flight conditions are such that the rotational temperature is comparable to the vibrational temperature in the 4-T model due to the extremely high translational temperature of over 40 000 K. In the high-temperature regime, the translational-rotational energy assumptions of the 2-T and 3-T models cannot be guaranteed. FIRE-II condition has similar altitude conditions as RAM-C II, but the high velocity and large nose radius result in a large thermochemical equilibrium regime.

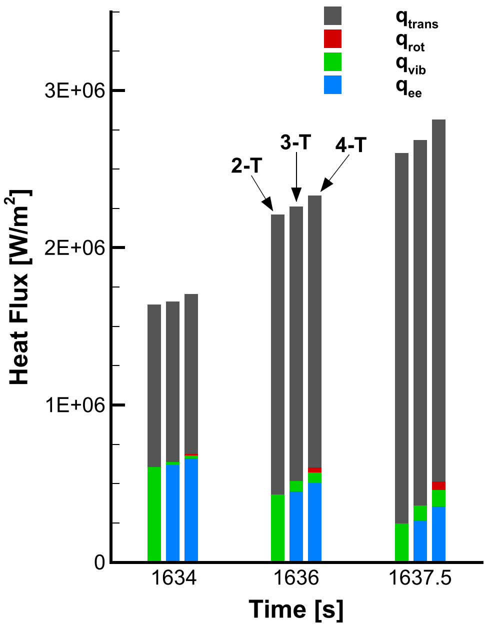


FIG. 14. Surface heat flux components for each energy mode at FIRE-II flight conditions.

Figure 16 compares the electron temperature and number density normal to the wall of the multi-T models at the shoulder for the 1634 and 1637.5 s conditions. Although the shoulder is in the supersonic regime past the sonic line of the bow shock, it is located about 21° from the nose and shows an electron temperature similar to the stagnation line. The electron temperature is different for each model immediately after the shock wave, but the electron number density of 3-T model has similar distribution as that of the 2-T model. The electron temperature and number density of the 4-T model increased earlier due to the increase in shock standoff distance, but it converged to the similar level as the 2-T and 3-T models near the maximum electron temperature. The electron number density near the wall of the multi-T models changed at a rate similar to that of the surface heat flux increase. FIRE-II showed a similar flow structure throughout the forebody with a large nose radius and high velocity. In the large equilibrium regime, the temperature distribution and electron number density were similar among the multi-T models.

E. LENS-XX: Double cone, hollow cylinder-flare

Double cone and hollow cylinder-flare experiments[44] were conducted to analyze the SWBLI phenomenon in high enthalpy flow at CUBRC's LENS-XX expansion tunnel,[69] which is the identical to the previous cylinder. In particular, the peak pressure, peak heat flux, and separation region generated by the shock–shock interaction of the double cone are challenging to predict accurately by numerical simulation, so various studies have been performed to analyze the SWBLI phenomenon.[58,70–77] Numerical simulations of double cone and hollow cylinder-flare were performed to verify the effects of multi-T models on surface heat flux and electrons in the SWBLI phenomenon. The double cone is composed of two cones of 25° and 55° with a length of about 0.19 m, and the hollow cylinder-flare is composed of a hollow cylinder of 65.024 mm diameter and a flare of 30° with a length of about 0.22 m. The free stream conditions of the double cone and hollow cylinder-flare simulated are listed in Table IX. The wall boundary conditions are solved only for the non-catalytic case due to the minor influence of the catalytic wall.[73] The experimental uncertainty reported from the literature is 10%.[72]

Figure 17 compares the surface heat flux of the double cone of multi-T models and other numerical studies[70,71,75] with experimental data.[44] The multi-T models predicted the heat flux distribution and separation length in good agreement with other numerical studies in all free stream conditions. The separation region of the double cone is generated by the adverse pressure gradient by the interaction of the oblique shock of a 25° cone and the bow shock of a 55° cone.[70,71,73]

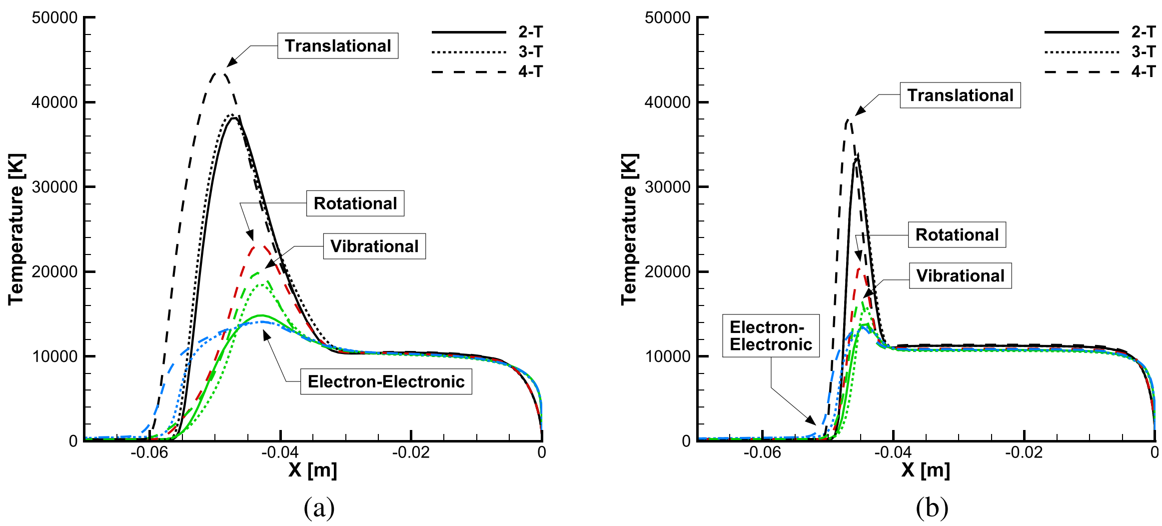


FIG. 15. Temperature distribution along stagnation line of multi-T models at FIRE-II flight conditions: (a) 1634 and (b) 1637.5 s.

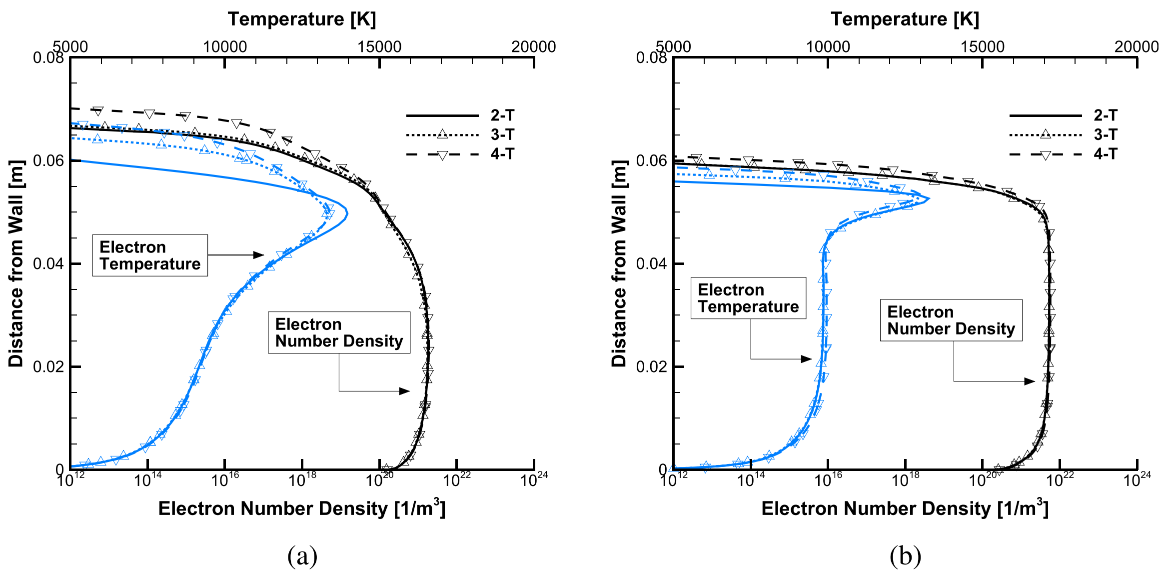


FIG. 16. Electron temperature and number density normal to the wall at the shoulder of FIRE-II flight conditions: (a) 1634 and (b) 1637.5 s.

The interaction of the separation shock and the bow shock generates a triple point, and the interaction of the triple point and the separation region makes it challenging to predict the separation region accurately.[70] The 4-T model increased the shock standoff distance of the bow shock in the blunt nose case. However, the influence is not significant near the triple point, and the differences in separation length and heat flux distribution among the multi-T models are negligible.

The vibrational temperature, electron-electronic temperature, and streamline of the 3-T model around the triple point of double cone Run 4 are compared in Fig. 18. Near the triple point, bow shock and supersonic jet occur. The supersonic jet passes through the oblique shock and separates the gas compressed by the bow shock with a relatively low temperature from the surface, reducing the surface heat flux.[73] The high flow velocity and low temperature create conditions close to thermochemical frozen, resulting in a nonequilibrium regime

TABLE IX. Free stream condition of LENS-XX double cone and hollow cylinder-flare experiment.

Run	ρ_∞ (kg/m^3)	T_∞ (K)	T_{wall} (K)	U (m/s)	Test gas	Model
1	0.499×10^{-3}	175	300	3246	Air	DC
2	0.984×10^{-3}	389	300	4303	Air	DC
3	0.510×10^{-3}	521	300	6028	Air	DC
4	0.964×10^{-3}	652	300	6497	Air	DC
5	1.057×10^{-3}	523	300	5996	Air	DC
6	2.045×10^{-3}	573	300	5466	Air	DC
1	0.634×10^{-3}	189	300	3123	Air	HCF
2	0.499×10^{-3}	318	300	4497	Air	HCF
3	1.750×10^{-3}	383	300	4660	Air	HCF
4	2.216×10^{-3}	569	300	5470	Air	HCF
5	0.947×10^{-3}	618	300	6515	Air	HCF

for vibrational and electron-electronic temperatures. Although the double cone flow structure results in thermal nonequilibrium of the supersonic jet near the surface, the electron-electronic heat flux was so small that its effect on the total heat flux is negligible.

The electron temperature and number density normal to the wall at 30% from the junction of the 55° cone are compared in Fig. 19 to determine the impact of the multi-T models on the supersonic jet. The supersonic jet is located about 1.5 mm away from the wall, predicting electron temperatures and number density at about half of those predicted by the 3-T and 4-T models. The impact of the 4-T model on electrons is comparable to that of the 3-T model. In Run 3, 4, 5, and 6, where the total enthalpy and degree of ionization are high, the electron temperature and number density exhibit a similar trend. In Run 1 and 2, however, where the degree of ionization was small, there are negligible changes in electron number density between the multi-T models.

The hollow cylinder-flare model has a shock–shock interaction similar to the double cone, which makes it difficult to predict the separation length and accurate heat flux distribution. However, the surface heat flux and degree of ionization are minimal due to weak shock, even in free stream condition similar to double cone model. Figure 20 compares the surface heat flux of the multi-T models of the hollow cylinder-flare and other numerical studies[70,78–80] with the experimental data.[44] In Run 1, 2, and 5, no separation occurred due to the low density, and all multi-T models accurately predict the peak heat flux and narrow separation length within error bars of the experimental data. The separation length of the 4-T model increases compared to the 2-T and 3-T models, and the position of the peak heat flux shifted, though the change is negligible.

Run 3 and 4 are free stream conditions with similar total enthalpy to Run 2, but with 3.5 and 4.4 times larger density, respectively, resulting in a wide separation region.[70] The hollow cylinder-flare model exhibits a simpler flow structure than the double cone model. Furthermore, the correlation between the separation length and the location of the peak heat flux due to the shock–shock interaction is

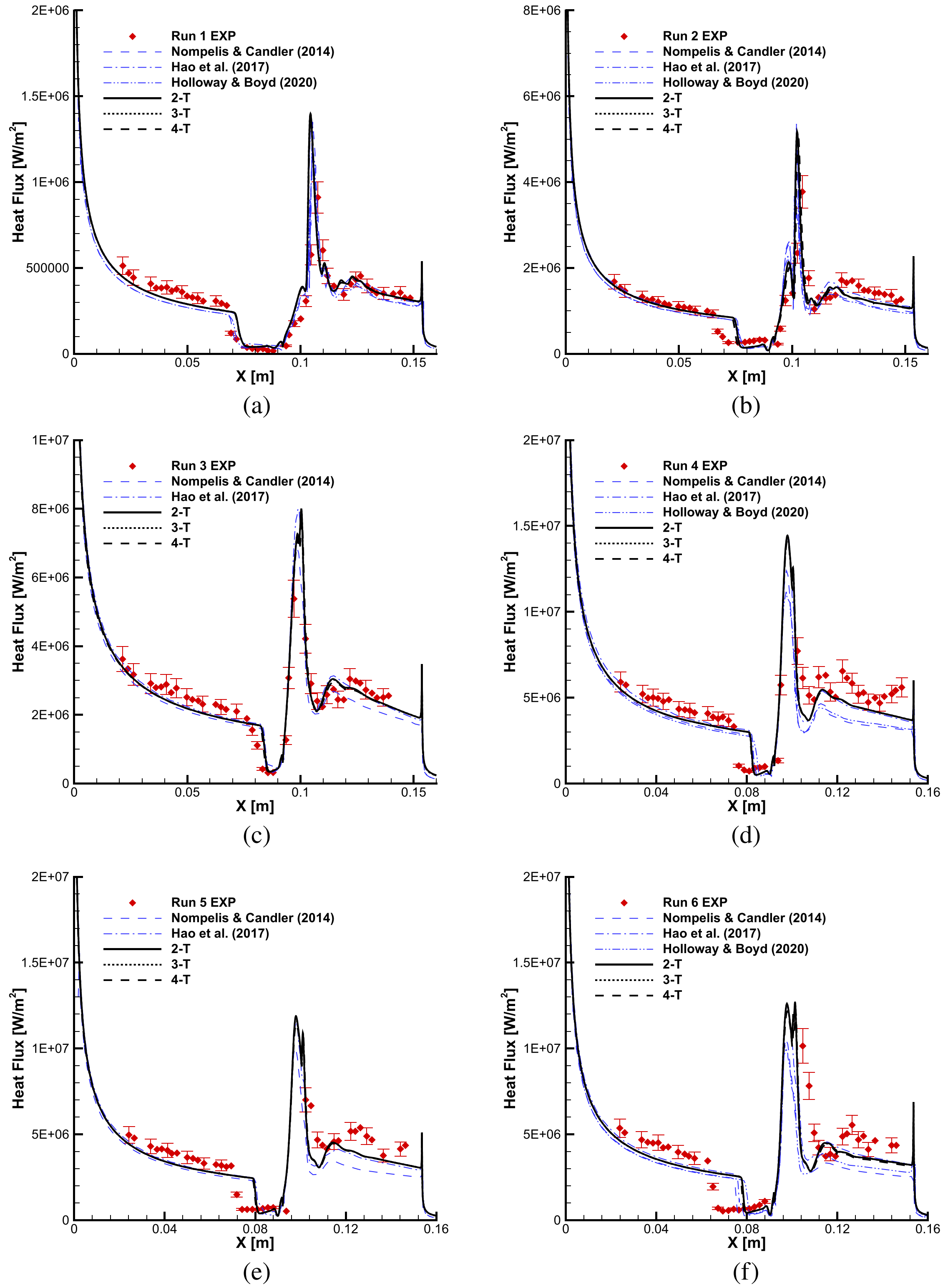


FIG. 17. Comparison of surface heat flux of LENS-XX double cone model with experiment data[44] and other numerical studies:[70,71,75] (a) Run 1, (b) Run 2, (c) Run 3, (d) Run 4, (e) Run 5, and (f) Run 6.

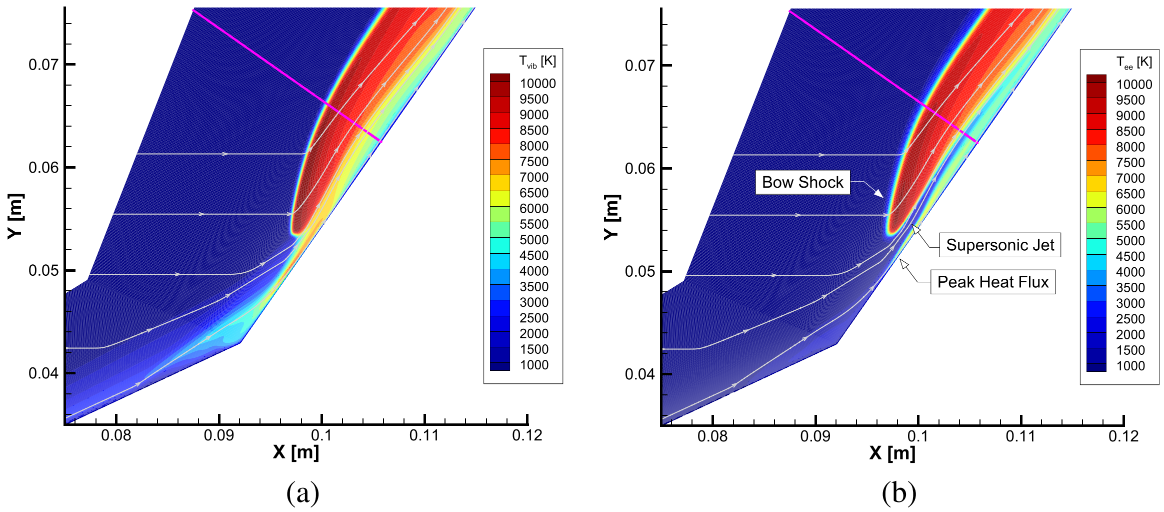


FIG. 18. Temperature distribution of 3-T model around triple point of LENS-XX double cone Run 4: (a) Vibrational temperature and (b) electron-electronic temperature.

clear. The location of the peak heat flux shifts downstream as the separation length increases. The multi-T models overpredict the separation length compared to the experimental data in Run 3 and 4. The peak heat flux also lies outside of the error bars, but the results are consistent with those observed in other numerical studies. In the 4-T model, the separation length is increased by 3.3% and 4.5% compared to the 2-T model, respectively, and the location of the peak heat flux is shifted by about 2.6% and 3.4% to the downstream, but the peak heat flux remains similar.

The electron temperature and number density distributions of the multi-T models for the hollow cylinder-flare Run 4 are compared

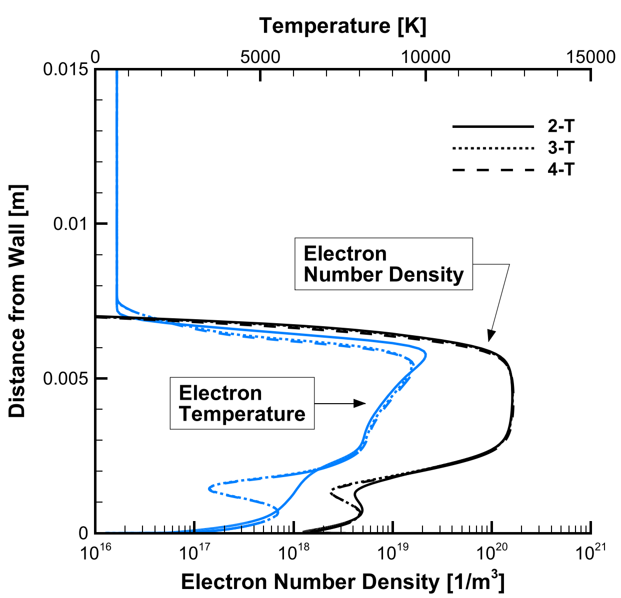


FIG. 19. Electron temperature and number density of multi-T models normal to wall at 30% of 55° cone from junction of LENS-XX double cone Run 4.

in Fig. 21. Run 4 is not the largest total enthalpy condition among the SWBLI cases, but it has the largest surface heat flux due to the separation shock. In the separation region, the hollow cylinder-flare model is comparable to the double cone, resulting in electron temperature differences of several thousand Kelvin among the multi-T models. However, the difference in electron number density is relatively minor due to the low degree of ionization. In the hollow cylinder-flare model, a weak correlation between electron temperature and number density is observed due to the low degree of ionization.

IV. CONCLUSION

Thermochemical nonequilibrium multi-T models were developed by separating the energy integrated in the existing 2-T model. Numerical simulations were performed under RAM-C II, Electre, LENS-XX experiments, and FIRE-II conditions to determine the effects on surface heat flux and electrons in various flow phenomena. The 3-T model separates the electron-electronic energy from the electron-electronic-vibrational energy, and the 4-T model separates the rotational energy from the translational-rotational energy. For the energy relaxation transfer between the separated energy modes, the relaxation time models of Lee[16] and Laporta *et al.*[18,19] were applied for $e-vib$, and the model of Jo *et al.*[26] was applied for $trans-rot$.

In the thermochemical nonequilibrium 3-T model, significant changes in electron density and electron-electronic temperature were observed. Comparing the electron production rate along the stagnation line where the bow shock occurs revealed that changes in the electron temperature affected the distribution of electron density. However, despite differences in electron production rate compared to the 2-T model, the maximum electron density showed similar values. When the flow expands, the 3-T model predicts higher electron temperatures than the 2-T model due to the longer relaxation time of electron-electronic energy. In the SWBLI of the double cone model, the lower temperature and degree of ionization of the supersonic jet reduced the electron temperature and density. The difference in surface heat flux between the 2-T and 3-T models is negligible except for the FIRE-II flight test conditions. The influence of surface heat flux due to

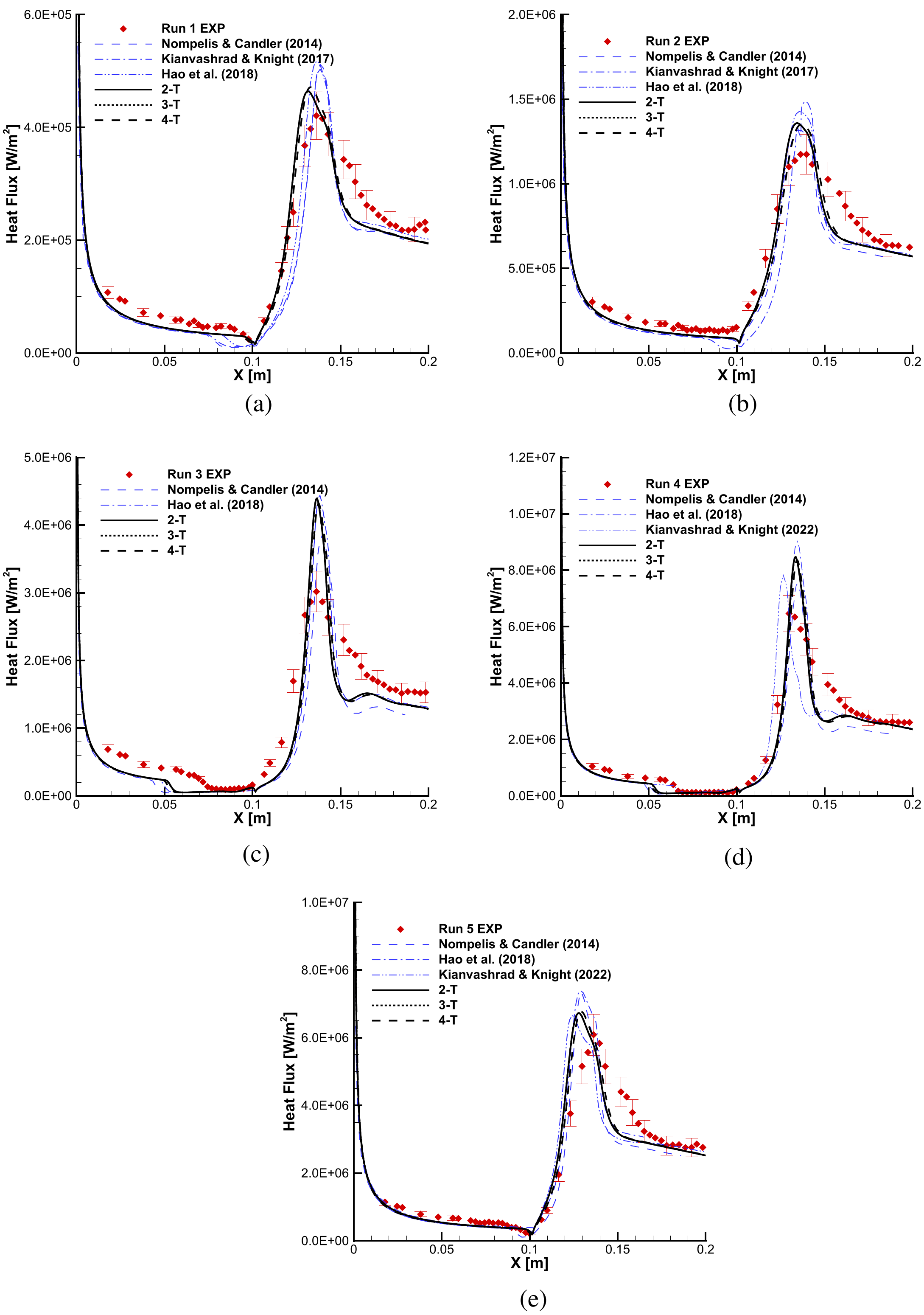


FIG. 20. Comparison of surface heat flux of LENS-XX hollow cylinder-flare model with experiment data[44] and other numerical studies:[70,78–80] (a) Run 1, (b) Run 2, (c) Run 3, (d) Run 4, and (e) Run 5.

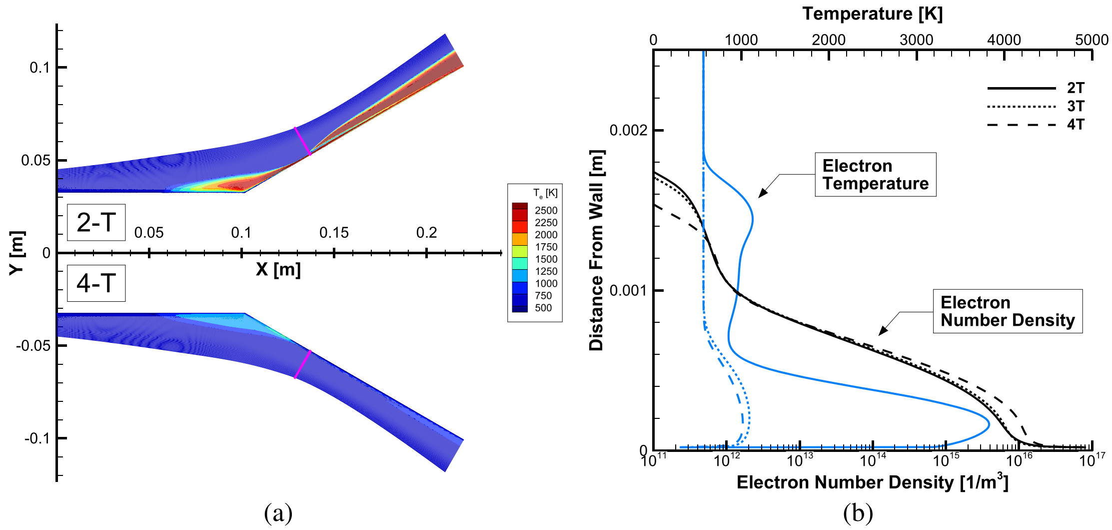


FIG. 21. Comparison of electron temperature and number density of multi-T models of LENS-XX hollow cylinder-flare Run 4: (a) Electron temperature distribution of 2-T and 4-T models, and (b) electron temperature and number density normal to wall at 30% from junction of flare.

electron-electronic nonequilibrium was significant only under conditions where the degree of ionization and electron-electronic thermal conductivity were sufficiently large.

In the thermochemical nonequilibrium 4-T model, pronounced changes due to the increase in shock standoff distance were observed. When comparing along the stagnation line properties among the multi-T models, the 4-T model showed earlier increases in electron temperature and density compared to the 2-T and 3-T models as the shock standoff distance increased. Although the shock standoff distance increased in the 4-T model, the location where thermochemical equilibrium was reached remained similar across the multi-T models, leading to an increase in the thermochemical nonequilibrium region in the 4-T model. Excluding the effect of the shock standoff distance, the 4-T model also showed similar electron temperature and density distributions to the 3-T model due to the electron-electronic nonequilibrium. The increase in shock standoff distance in the 4-T model changed the shock wave structure in the SWBLI phenomenon, predicting a larger separation length compared to the 2-T and 3-T models.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Chanho Kim: Conceptualization (equal); Methodology (equal); Software (equal); Visualization (lead); Writing – original draft (lead); Writing – review & editing (equal). **Kyu Hong Kim:** Conceptualization (equal); Methodology (equal); Validation (equal); Writing – review & editing (equal). **Yosheph Yang:** Formal analysis (equal); Methodology (equal); Software (equal); Writing – review & editing (equal). **Jae Gang Kim:** Conceptualization (equal); Methodology (equal); Project administration (equal); Software (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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